Chapter 14 — Case Studies: Everyday Objects (Study Edition)

> **Where this fits.** Chapters 1–13 gave you the tools (OOP, UML, SOLID, clean code, the GoF patterns, concurrency). Chapters 14–21 apply them to real interview questions. This chapter starts with **machines you have used yourself** — parking lot, vending machine, coffee machine, ATM, elevator, traffic light, parcel locker — so all your attention can go to the design. This is the study edition: every topic stays, the padding goes. The Full Edition (README.md) has the complete code, outputs and stage-by-stage course diagrams.

#	Problem	The one idea the interviewer is really testing	Main patterns
14.1	Parking Lot	**Resource allocation** under concurrency: two cars must never get the same spot	Strategy (spot finding, pricing), Factory, Observer (display boards)
14.2	Vending Machine	A **state machine**: the order in which actions are allowed	State
14.3	Coffee Vending Machine	**Atomic, all-or-nothing** deduction from a shared inventory; recipes as data	Observer (low-stock alerts), data-driven recipes, sealed result types
14.4	ATM	**Security + money correctness**: authenticate, never lose money, never double-spend	State, Chain of Responsibility (cash dispenser), Command (transactions)
14.5	Elevator System	**Scheduling** (SCAN/LOOK) and **dispatching** many cars	Strategy (dispatch), State-like modes
14.6	Traffic Signal Control	**Timed state machine** with a hard safety invariant and emergency override	State, Strategy (timing), Observer (safety monitor)
14.7	Shipping Locker System (Amazon Locker)	**Best-fit allocation + deadlines + a secret**: reserve a door atomically, expire holds and uncollected parcels, single-use hashed pickup codes	Strategy (door choice), Observer (SMS), enum state machines, sealed results

What these problems test

1. **State machines** — a fixed set of states plus transition rules: no dispensing before payment, no withdrawal before the PIN, never green on two crossing roads. The **State pattern** (Chapter 11) gives every button an answer in every state.
2. **Resource allocation** — spots, notes, milk, cars, locker doors handed out never twice, never below zero, fairly.
3. **Concurrency** — several gates, two ATMs on one account, a timer plus an emergency thread; the design must stay correct when calls overlap (Chapters 12–13).

> **Key insight.** The hardest part of each problem is **one method**: pick a spot, make change, deduct ingredients, pick the next stop. Find it early, name it out loud, spend your time there.

How each case study is laid out

Problem & clarifying questions → Requirements → Use cases → Core entities → Class diagram → Key flows → Design decisions & patterns → Java implementation → Concurrency & edge cases → Extensions & follow-ups → In the interview → Self-check → TL;DR. This is also a good order to speak in during an interview.

How to read the course diagrams

The course draws class diagrams in stages (earlier classes in grey, new ones in colour). This study edition keeps only the **finished** course diagram per case plus a mermaid diagram of our Java code.

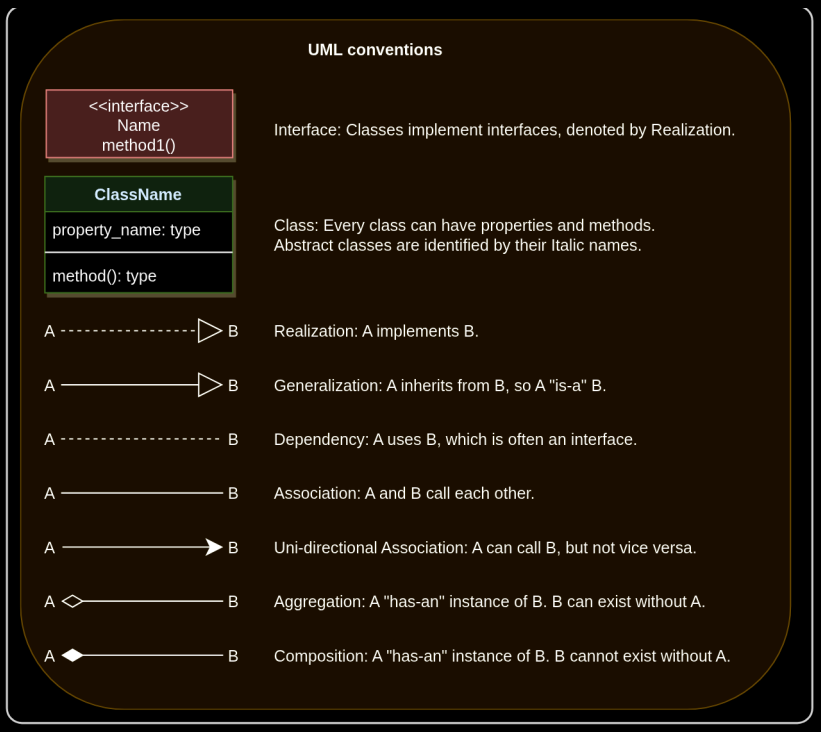


Figure — Legend for the course class diagrams: interface, class, realization, association, inheritance, composition and the other arrow kinds (course diagram)

Hollow triangle = inheritance ("is a"); **dashed line + hollow triangle** = implements an interface; **filled diamond** = composition (the part dies with the whole); **plain arrow** = association (refers to, does not own). See Chapter 3.

How to run every example in this chapter

```
cd chapters/14-case-everyday-objects/code
javac --release 21 -d /tmp/lld_14 $(find . -name '*.java')
java -cp /tmp/lld_14 lld.ch14.parkinglot.Main      # or vendingmachine, coffeemachine, atm, elevator,
trafficsignal, locker
```

Java 21, standard library only. **Code-style note:** some files hold several small package-private types (e.g. `Vehicles.java`); fine except for the `auxiliaryclass` lint — in production give each type its own file.

14.1 Design a Parking Lot

A **parking lot** (car park) with many floors, many spot sizes and several ways in and out. It is the **most-asked LLD question**. It looks easy, so the interviewer can watch how you **scope** it, how you model the **fit rule** (which vehicle may use which spot), and whether you notice that **two gates can hand out the same spot** at the same moment.

Problem & clarifying questions

Question	Assumed answer
Scope?	One lot, many floors; several entrances and exits working at once.
Vehicles / spots?	Car, truck, van, motorcycle, EV / compact, large, handicapped, motorcycle, electric.
Payment?	Cash or card at the exit panel, an attendant, or a per-floor info portal.
Fee?	$4 first hour, $3.50 hours 2–3, $2.50 after; a started hour counts as full.
EV charging?	Electric spots have a pay-and-charge panel.
Reservations, passes, plate recognition?	Out of scope (extensions).

Requirements

Functional: (1) multiple floors; (2) multiple entries/exits; (3) ticket at entry, pay at exit; (4) pay at the exit panel or an attendant; (5) cash or card; (6) pay early at a per-floor **info portal** (not charged again); (7) **never exceed capacity** — "full" on the entrance panel and ground-floor board; (8) spot types; (9) **electric** spots with a pay-and-charge panel; (10) vehicle types; (11) a **display board per floor**; (12) tiered hourly fees.

Non-functional: correct under concurrency (no shared spot, no double payment); fast gate (O(log n) from a sorted free set); extensible; testable (injectable clock).

Use cases

Actors: Admin (floors, spots, attendants, rates), Customer, Parking attendant (a customer proxy who takes cash), System (messages, assigns/frees spots). **Top use cases: take ticket**, **scan ticket**, card/cash payment, admin edits, set rate.

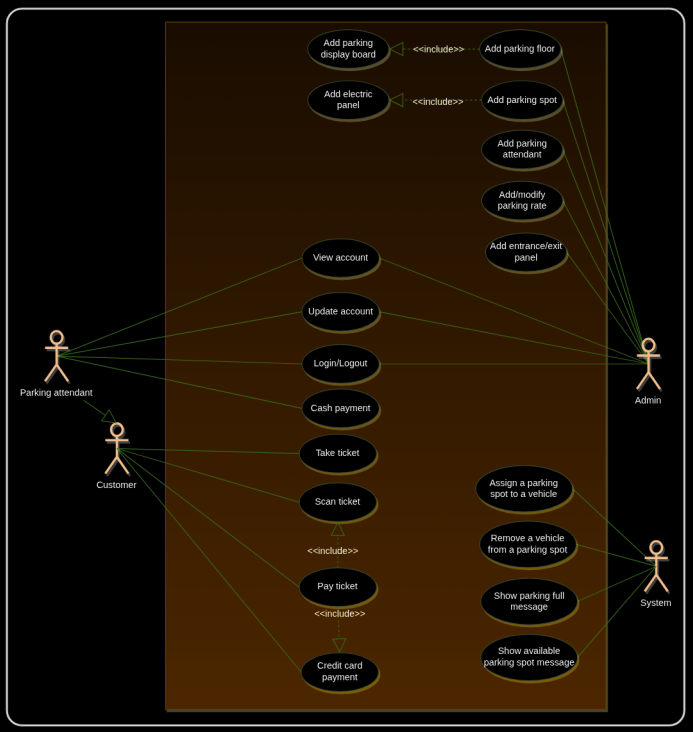


Figure — Use case diagram of the parking lot: Admin, Customer, Parking attendant and System, with their use cases (course diagram)

Core entities

Class	Responsibility
`VehicleType` (enum)	Kinds of vehicle and, for each, the spot types it fits, **best fit first**.
`SpotType` (enum)	Motorcycle, compact, large, handicapped, electric.
`Vehicle` (record)	Plate, type, accessibility permit; knows its preferred spot types.
`ParkingSpot`	One spot: id, floor, type, occupant (atomic).
`ChargingPanel`	Pay-and-charge panel owned by an electric spot.
`SpotFactory`	Builds spots; the only place that knows an electric spot needs a panel.
`ParkingFloor`	Free spots per type in a thread-safe sorted set; notifies boards.
`DisplayBoard`	Free spots per type on one floor (an observer).
`SpotAllocationStrategy`	*How* to choose a spot.
`PricingStrategy`	*How much* a stay costs.
`Ticket`	Vehicle, the **actual spot**, entry time, status ACTIVE → PAID → EXITED.
`PaymentMethod`	Cash or card.
`ParkingLot`	Ties it together: admit, price, pay, exit.
`EntrancePanel`, `ExitPanel`, `CustomerInfoPortal`	Devices customers touch; **no rules**, they call the lot.

The course also has `Account` with `Admin` and `ParkingAttendant` subclasses; they carry no parking logic, so the code omits them.

Class diagram

The course grows the design in **five stages**, each leaving something that runs (a good interview habit: if time runs out, what you drew already works):

Stage	Adds	Key decision
1	Vehicle and spot types	Spot holds **0..1** vehicle; the **fit rule lives on the vehicle** (truck/van → large; EV → electric, compact, large).
2	Floors, boards	Floor owns spots and board; board stores **counts**, not a stale "sample free spot"; one map keyed by type.
3	Ticket, rate, payment	**Ticket stores the exact spot**; abstract `Payment` with cash/card (LSP holds).
4	`ParkingLot`	Owns floors and rate; course makes it a **Singleton** and serialises assignment.
5	Panels, portal, electric panel	Panels are **thin**, forwarding to the lot; electric panel owned by its spot.

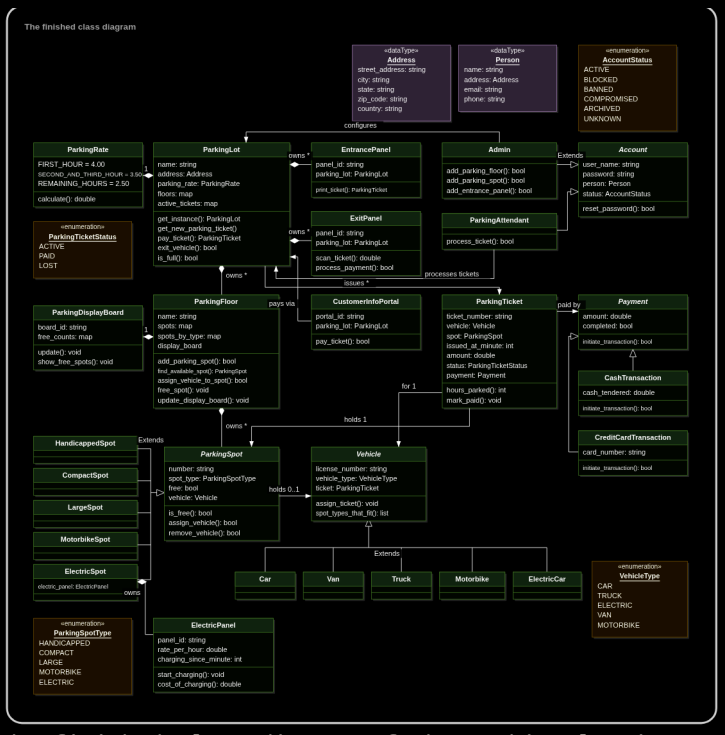
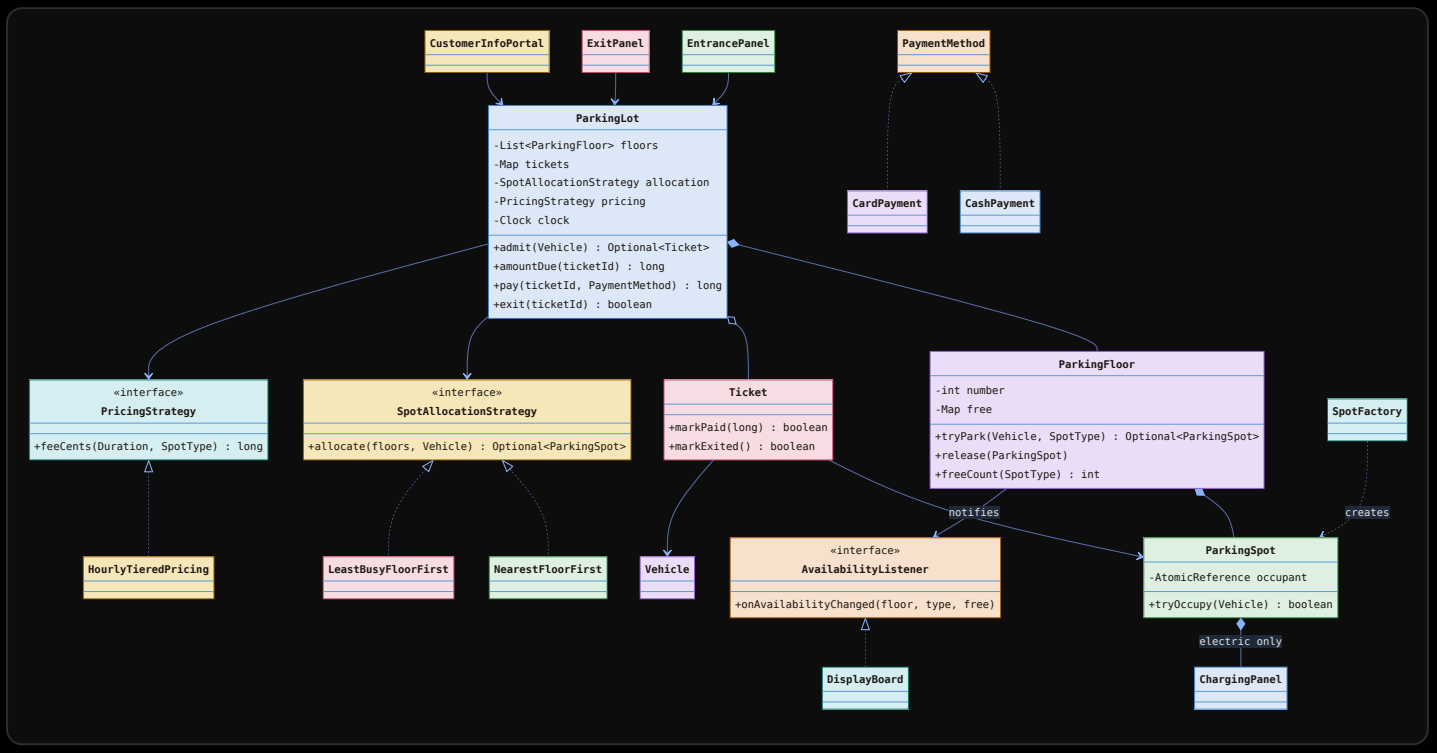


Figure — The finished class diagram of the parking lot (course diagram)

Our Java code keeps the same shape with three changes interviewers push for: spot types are an **enum** (not five empty subclasses), pricing and spot-finding are **Strategies**, and the display board is an **Observer**.



Key flows

- **Entry:** `EntrancePanel` → `lot.admit(vehicle)` → strategy tries each preferred spot type on each floor → `floor.tryPark` does `pollFirst()` on the free set (**atomic**) and notifies the board → ticket printed, or "LOT FULL".
- **Exit:** `ExitPanel` → `lot.pay(ticketId, card)` **locks the ticket**; if ACTIVE, price it and charge, then mark PAID; if already PAID at the portal, charge 0 → `lot.exit` moves PAID → EXITED and releases the spot → barrier opens.
- **Ticket life cycle:** `ACTIVE → PAID → EXITED`.

Design decisions & patterns used

Decision	Pattern / principle	Why	Alternative rejected
Fit rule on `VehicleType`	Information Expert, OCP	New vehicle type = one enum constant	`switch(vehicleType)` in the lot
Spot type is an **enum field**	Composition over inheritance	The 5 course subclasses had no behaviour	`CompactSpot`, `LargeSpot`…
`SpotAllocationStrategy`	**Strategy**	Nearest floor, least busy, VIP… swap freely	`if (mode == NEAREST)`
`PricingStrategy`	**Strategy**	"What if pricing changes?"; admin swaps at run time	Tiers hard-coded in `Ticket`
`SpotFactory` attaches `ChargingPanel`	**Factory**	One place knows "electric ⇒ has a panel"	Every caller remembers
Boards subscribe to floors	**Observer**	Boards, apps, bay lights update without the floor knowing them	Lot pushing into boards
Ticket holds the **spot**	—	Exit frees the exact spot	Counting occupied spots
Lot is **not** a Singleton	Dependency injection	A Singleton is a hidden global; tests can't build two lots	`ParkingLot.getInstance()`
`Clock` injected	Testability	Demo jumps hours instantly	`Instant.now()` inside

> **Honest note on "Strategy".** A cash/card subclass pair is just an abstract base class; Strategy means an object **holds** a swappable behaviour (`setPricing(...)`). The course admits the lot does two jobs; spot-finding moved to strategy and floors, and tickets could move to a `TicketRepository`.

Java implementation

Package `lld.ch14.parkinglot`. Amounts are **cents** (`long`) to avoid floating-point rounding.

```
enum VehicleType {
    MOTORCYCLE(List.of(SpotType.MOTORCYCLE)),
    CAR(List.of(SpotType.COMPACT, SpotType.LARGE)),
    ELECTRIC_CAR(List.of(SpotType.ELECTRIC, SpotType.COMPACT, SpotType.LARGE)),
    VAN(List.of(SpotType.LARGE)),
    TRUCK(List.of(SpotType.LARGE));
    // ...
}

class ParkingFloor {
    private final Map<SpotType, ConcurrentSkipListSet<ParkingSpot>> free = new EnumMap<>(SpotType.class);
    // ...
    /** pollFirst() is atomic: two threads can never receive the same spot. Lowest id = nearest. */
    Optional<ParkingSpot> tryPark(Vehicle v, SpotType type) {
        ParkingSpot spot = free.get(type).pollFirst();
        if (spot == null) return Optional.empty();
        if (!spot.tryOccupy(v)) throw new IllegalStateException("free-set out of sync for " + spot);
        notifyListeners(type);
        return Optional.of(spot);
    }
    // ...
}

final class ParkingLot {
    // ...
    Optional<Ticket> admit(Vehicle v) {
        String placeholder = "pending";
        if (ticketByPlate.putIfAbsent(v.plate(), placeholder) != null) {
            throw new IllegalStateException(v.plate() + " is already inside");
        }
        Optional<ParkingSpot> spot = allocation.allocate(floors, v);
        if (spot.isEmpty()) {
            ticketByPlate.remove(v.plate());
            return Optional.empty();
        }
        Ticket t = new Ticket("T" + nextTicket.getAndIncrement(), v, spot.get(), clock.instant());
        tickets.put(t.id(), t);
        ticketByPlate.put(v.plate(), t.id());
        return Optional.of(t);
    }
    // ...
    long pay(String ticketId, PaymentMethod method) {
        Ticket t = ticket(ticketId);
        synchronized (t) {                // per-ticket lock: portal and exit panel cannot both charge
            long due = amountDue(ticketId);
            if (due == 0) return 0;
            method.pay(due);              // throws if the payment fails: ticket stays ACTIVE
            t.markPaid(due);
            return due;
        }
    }
    // ... exit(): markExited() (PAID -> EXITED), then floor.release(spot)
}
```

Full runnable code: code/lld/ch14/parkinglot/ (see the Full Edition).

How it works: `Vehicle.preferredSpots()` puts `HANDICAPPED` first for permit holders only. `ParkingSpot.tryOccupy` (`compareAndSet(null, v)`) is a second safety net. `NearestFloorFirst` and `LeastBusyFloorFirst` are interchangeable strategies.

Output (excerpt):

```
[entry-A] TRUCK-2 -> SORRY, LOT FULL for TRUCK
[portal] T2 due $16.00, charged $16.00
[exit-A] T2: already paid, barrier OPEN
Unpaid exit attempt: barrier opens? false
== 20 cars race for the remaining CAR spots ==
cars admitted: 5, distinct spots: 5
```

4 h 50 min rounds to 5 hours = $4 + $3.50 + $3.50 + $2.50 + $2.50 = **$16.00**; 20 racing threads get exactly 5 distinct spots.

Concurrency & edge cases

Situation	Risk	Handling
Two gates, last spot	Double booking	Atomic `pollFirst()` + CAS `tryOccupy`
Portal and exit pay the same ticket	Charged twice	`synchronized (ticket)`; a PAID ticket costs 0
Same plate enters twice	Two tickets	`ticketByPlate.putIfAbsent`
Card declined	Marked paid anyway	`method.pay` throws **before** `markPaid`
Leaves without paying	Fee lost	`exit` only from PAID; barrier stays closed
Pays at portal, stays 3 more hours	Free parking	**Grace period** (e.g. 15 min); add `paidAt`, re-price after
Board shows an old number	Out-of-order notifications	OK for display; board can **pull** `freeCount` on render
Lost ticket / 0-minute stay	No entry time / $0	Daily maximum or plate lookup / minimum one hour
Lock scope	Global lock serialises gates	Per-spot CAS and per-ticket lock only

`ConcurrentSkipListSet.size()` is not constant-time and only an estimate under change — fine for a board, never for deciding a spot is free (use `pollFirst()`).

Extensions & follow-up questions

1. **Pricing changes** (weekend, per spot type, surge) → new `PricingStrategy` (or a decorator), `setPricing`. Nothing else changes.
2. **Reservations** → `Reservation(plate, spotType, from, to)` + reserved set per floor; hold with a timeout (e.g. 30 min).
3. **Monthly passes** → pricing returning 0 for pass plates, or a `PassHolderPricing` decorator.
4. **Plate recognition** → the camera is another entrance panel calling `admit`.
5. **Database** → tables `floor`, `spot`, `ticket`, `payment`; claim with `UPDATE spot SET occupied_by=? WHERE id=? AND occupied_by IS NULL` (the DB equivalent of CAS). Show **cardinalities** (floor 1..* spots, spot 0..1 vehicle, ticket 1 spot).
6. **Many lots in a city** → a `ParkingNetwork` routing to lots — another reason for no Singleton.
7. **EV idle fees** → a charging session on `ChargingPanel` with a per-minute fee after charging completes.

Real-world example

Parking guidance systems (per-bay sensors, free counts per level) are our `DisplayBoard` observers; airports use **plate recognition**; SpotHero/ParkMobile pre-booking = reservations; Tesla **idle fees** = the EV extension.

In the interview

45 minutes: 0–5 questions/requirements → 5–10 use cases → 10–25 stages 1–4 ("fit rule on the vehicle", ticket holds the spot) → 25–38 code `admit`, `tryPark`, `pay`, `exit` → 38–45 concurrency and extensions.

> **30-second answer:** "Vehicles and spot types are enums with the fit rule on the vehicle; floors keep a concurrent sorted set of free spots per type; the lot admits, prices and releases. Tickets point at the real spot. Pricing and spot-finding are Strategies. Handing out a spot is atomic — `pollFirst` plus a CAS on the spot — so two gates never get the same spot, and payment locks only the one ticket."

- *"Why not a Singleton?"* — It is a hidden global that makes testing two lots impossible; I create one and inject it. If pressed, a lazy holder-class Singleton is thread-safe.
- *"How do you make finding a spot fast?"* — Keep only free spots, indexed by type, in a sorted set: O(log n), no scan.

Self-check

1. **Q:** Why must the ticket store the spot? **A:** So the exit frees exactly that spot; a count can't say which.
2. **Q:** What makes assignment thread-safe? **A:** `pollFirst()` gives a spot to exactly one caller; `compareAndSet` double-checks.
3. **Q:** Where does a weekend price go? **A:** A new `PricingStrategy` via `setPricing`.
4. **Q:** Paid at the portal — what does the exit charge? **A:** Nothing: `amountDue` is 0 for PAID; `exit` moves PAID → EXITED.

TL;DR

- Fit rule on the **vehicle**, spot type as an **enum**, free spots per type in a **concurrent sorted set**.
- **Ticket → spot** reference; status ACTIVE → PAID → EXITED.
- **Strategy** (pricing, spot-finding), **Factory** (electric spots), **Observer** (boards); inject the lot, no Singleton.
- Atomic `pollFirst` + CAS for spots, per-ticket lock for payment, `putIfAbsent` for plates.

14.2 Design a Vending Machine

A **vending machine** is an unattended shop: key in a slot code, pay, take the item from the tray. The interviewer skips the domain and checks one thing: **which actions are legal at which moment**. Accepting a coin with nothing chosen, or two items for one payment, are bugs no tidy code elsewhere can excuse.

Problem & clarifying questions

Question	Assumed answer
Select or pay first?	Select a slot code first, then pay.
Which money?	Fixed set: 5¢, 10¢, 25¢, $1, $5. Anything else is returned.
Card / contactless?	Out of scope (extension).
Change?	Yes. If **exact** change is impossible, refuse the sale and return everything.
Cancel?	Any time before dispensing, full refund.
Concurrency?	One customer at a time (one physical machine).
Restocking?	An operator restocks, loads coins, collects takings.

Requirements

Functional (the course's twelve): (1) many **slots**, each with a code, a **single** product and a **count**; (2) a product has a name and a **price in cents**; (3) **choose, then pay**; (4) only known coins/notes, others returned; (5) the **display** says what to do after every action; (6) when paid enough, release **item and change**; (7) if **exact change is impossible**, refuse and return every cent; (8) **cancel** with full refund until the item drops; (9) an empty slot **cannot be chosen**; (10) one payment = **exactly one** item; (11) an **operator** refills, adds change, removes takings; (12) item and cash counts are **exactly right** after every sale, cancel and refusal.

Non-functional: no illegal transitions (every button answered in every state); integer cents only; extensible (new payment device or state without editing every class).

Use cases

Actors: Customer, Operator, System. **Use cases:** select · insert money · dispense · collect · cancel · restock · load change / collect takings. "Dispense" is a **System** use case — the customer only causes it.

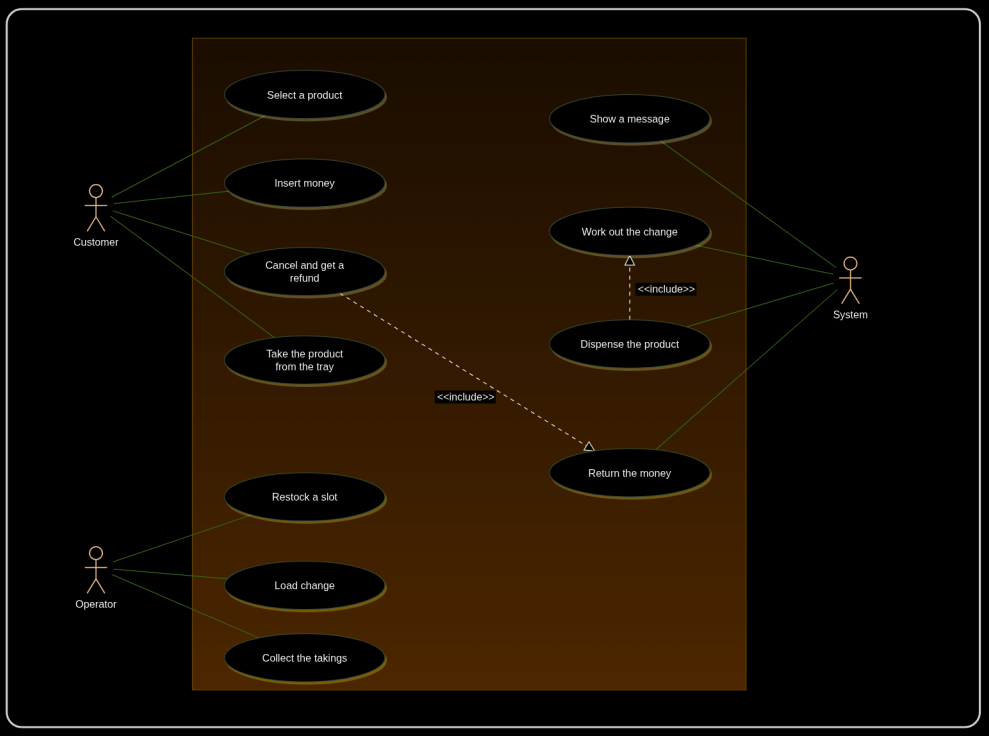


Figure — Use case diagram of the vending machine: Customer, Operator and System (course diagram)

Core entities

Class	Responsibility
`Product` (record)	Name and price in cents — *what* is sold.
`Slot`	One column: code, one product, **count** left.
`Inventory`	All slots by code.
`Denomination` (enum)	Accepted coins/notes and their cents.
`CashBox`	Counts pieces by value; plans **exact** change or reports it can't.
`VendingState` (interface)	One method per customer button: `select`, `insert`, `cancel`, `take`.
`IdleState`, `SelectedState`, `HasMoneyState`, `DispensingState`, `OutOfStockState`	The situations; each answers every button and names the next state (course calls `HasMoneyState` `PaidState`).
`Display`, `Keypad`, `CashSlot`, `Dispenser`, `Operator`	Course hardware/actor classes; in our code folded into `VendingMachine.show(...)` and its public/operator methods.
`VendingMachine`	Parts, current state, selection, balance; forwards each button to the state.

Why is the count on the slot, not the product? `Product` answers "what is it?"; `Slot` is the physical column. The same product can sit in two columns (even at different prices), and emptying one must not affect the other.

Class diagram

Course stages: (1) `Product`, `Slot` (association — a product can be in many slots), `Inventory` (composition); (2) `Coin`, `Note`, `CashBox` — **a piece goes into the box on insert**, so a refund is "pay exact change equal to the balance"; the planner returns an exact plan or "impossible"; (3) **the heart: one class per state**, each button a method on one interface — a state that forgets a button does not

compile; think of a **grid** of buttons × states where every cell needs an answer; (4) `Display` and `VendingMachine` — no `if` on state; (5) `Keypad`, `CashSlot`, `Dispenser`, `Operator` point **to** the machine; `take` finishes the sale; operator actions stay **off** the state interface.

The course draws four states; our code adds `OutOfStockState` so an empty machine refuses customer buttons until restocked. Cancel and "no exact change" end identically, so there is a **single** refund method.

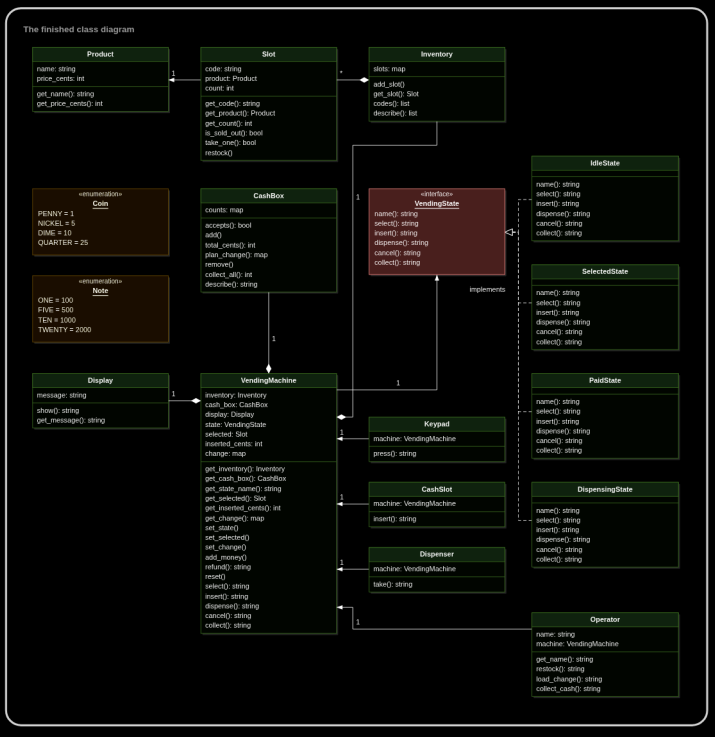
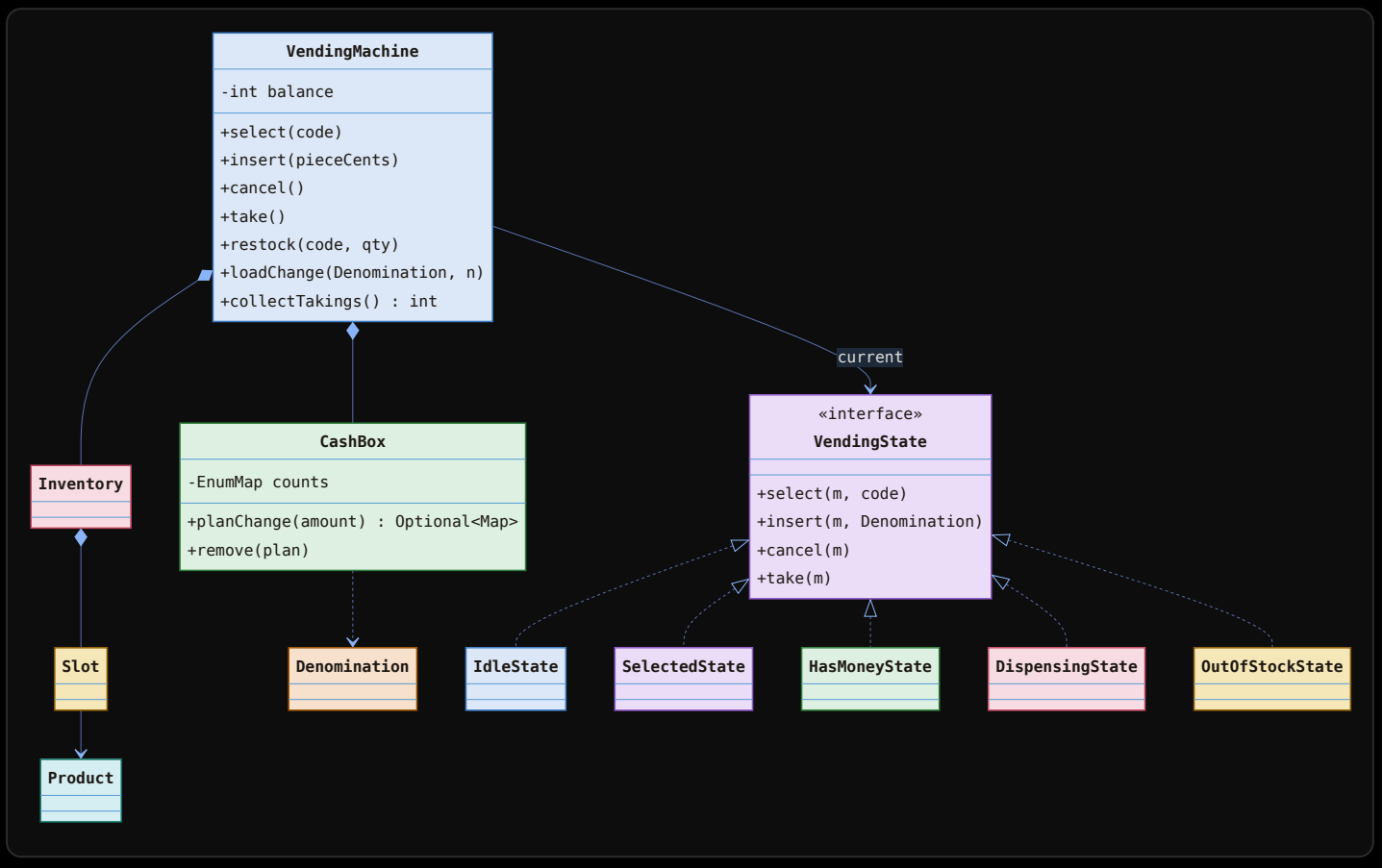
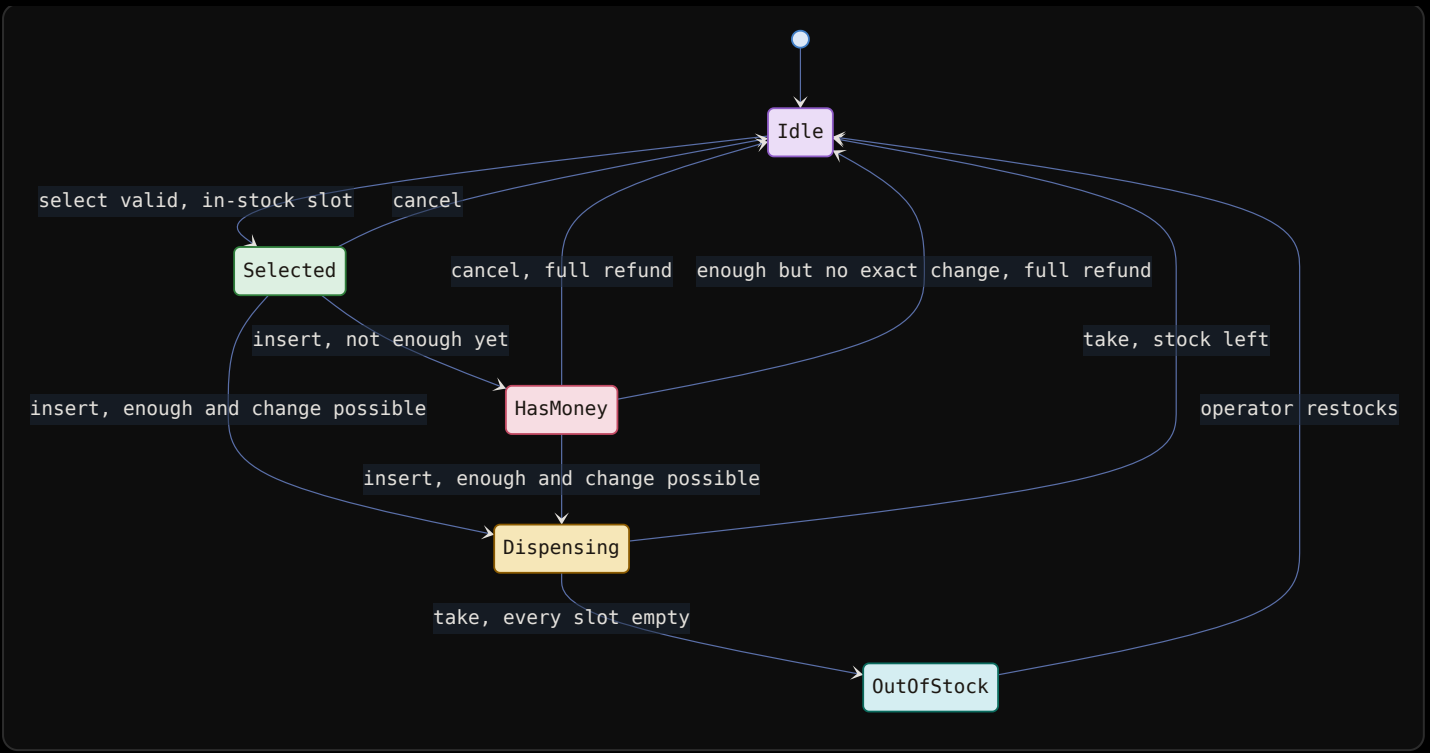


Figure — The finished class diagram of the vending machine (course diagram)



Key flows

The state machine is the design — draw it before coding.



State table (every cell answered):

Button ↓ / State →	Idle	Selected	HasMoney	Dispensing	OutOfStock
`select`	choose slot (reject empty/unknown)	change choice	"cancel first"	"take your product first"	"out of stock"
`insert`	return piece	accept; check paid	accept; check paid	return piece	return piece
`cancel`	"nothing to cancel"	clear choice → Idle	**refund all** → Idle	"too late"	"nothing to cancel"
`take`	"tray empty"	"pay first"	"pay the rest"	give product + change → Idle/OutOfStock	"tray empty"

Paying enough: insert → `acceptPiece` (into box, balance += value) → `checkPaid` → `planChange(balance − price)`; if a plan exists, remove it, `dispenseOne`, go to Dispensing; otherwise refund all and go Idle.

Design decisions & patterns used

Decision	Pattern / principle	Why	Alternative rejected
One class per situation	**State**	Every cell answered; compiler enforces completeness	Flags + `if/else` (double-dispense bugs)
States are stateless singletons (`INSTANCE`)	Flyweight-style sharing	No data, so one instance each	`new IdleState()` each transition
Operator actions off the state interface	Interface Segregation	Restock is valid in every state	`restock` on `VendingState`
Money counted by value in an `EnumMap`	—	Which physical quarter comes back doesn't matter	List of coin objects
Inserted money straight into the box	—	Refund = make change for the balance	Separate escrow pile
Exact change by dynamic programming	Correctness	Greedy fails when short of one coin	Greedy only
Refuse if exact change impossible	Req. 7	Never short-change	"Close enough" change

> **About the change algorithm.** The course's planner is **greedy** (largest piece first) and says so: correct for standard coin sets when well stocked, but it can fail when the box is short. The fix is **dynamic programming** — "coin change with limited coins". Example: 30¢ from {1 quarter, 3 dimes}: greedy takes the quarter and is stuck at 5¢; DP returns three dimes.

Java implementation

Package `lld.ch14.vendingmachine`.

```java
/** State pattern: every button is on the interface, so every state must answer every button. */
interface VendingState {
    void select(VendingMachine m, String code);
    void insert(VendingMachine m, Denomination d);
    void cancel(VendingMachine m);
    void take(VendingMachine m);
    String name();
}

/** Some money is in, but not enough yet. */
final class HasMoneyState implements VendingState {
    static final HasMoneyState INSTANCE = new HasMoneyState();

    @Override public void select(VendingMachine m, String code) { m.show("Cancel first to change your choice."); }
    @Override public void insert(VendingMachine m, Denomination d) { m.acceptPiece(d); m.checkPaid(); }
    @Override public void cancel(VendingMachine m) { m.refundAll("Cancelled."); }
    @Override public void take(VendingMachine m) { m.show("Please pay the rest first."); }
    @Override public String name() { return "HAS_MONEY"; }
}
// ... IdleState, SelectedState, DispensingState, OutOfStockState

final class VendingMachine {
    // ...
    /** Called after every accepted piece in SELECTED / HAS_MONEY. */
    void checkPaid() {
        int price = selected.product().priceCents();
        if (balance < price) {
            show("Paid %d c, %d c to go.".formatted(balance, price - balance));
            setState(HasMoneyState.INSTANCE);
            return;
        }
        var change = cashBox.planChange(balance - price);
        if (change.isEmpty()) {                              // refuse the sale rather than short-change
            refundAll("Cannot make exact change.");
            return;
        }
        cashBox.remove(change.get());
        tray.add(selected.dispenseOne().name());
        if (!change.get().isEmpty()) tray.add("change " + change.get());
        show("Dispensed. Please take: " + tray);
        balance = 0;
        selected = null;
        setState(DispensingState.INSTANCE);
    }

    /** One refund path for "cancel" and for "cannot make change". */
    void refundAll(String reason) {
        Map<Denomination, Integer> back = cashBox.planChange(balance)
                .orElseThrow(() -> new IllegalStateException("inserted pieces must be refundable"));
        cashBox.remove(back);
        show(reason + " Refunded " + balance + " c as " + back);
        balance = 0;
        selected = null;
        setState(IdleState.INSTANCE);
    }
    // ...
}
```

Full runnable code: code/lld/ch14/vendingmachine/ (see the Full Edition).

How it works: unknown pieces are returned before reaching a state. `CashBox.planChange` is bounded coin-change DP (fewest pieces); `greedyPlan` is only for comparison. `refundAll` cannot fail — the inserted pieces are in the box. Public methods are `synchronized`.

Output (excerpt):

```
 > insert 500c   (SELECTED)
   [display] Cannot make exact change. Refunded 500 c as {FIVE_DOLLAR=1}
30c from {1 quarter, 3 dimes}: greedy=Optional.empty dp=Optional[{DIME=3}]
```

Concurrency & edge cases

Edge case	Handling
Money before selecting	`IdleState.insert` returns it
Unknown coin / foreign note	Returned before reaching the state
Empty slot selected	Refused; stays Idle
Change choice after paying	"Cancel first" (avoids price confusion)
Overpayment	Change planned **before** dispensing; refuse if impossible
Cancel after dispensing	"Too late"
Two products for one payment	Impossible: balance reset to 0, Dispensing accepts no money
Last item sold	`take` → `OutOfStock`; only a restock leaves it
Two threads pressing buttons	`synchronized` public methods
Machines sharing one stock / power failure	Slot count needs a lock or atomic decrement / non-volatile transaction log (out of scope)

Extensions & follow-up questions

1. **Contactless card** → a new input device calling the same machine (e.g. `payByCard(amount)` setting the balance); states unchanged.
2. **Maintenance state** → a new `MaintenanceState` class; existing states untouched (OCP).
3. **Different price per slot** → move price from `Product` to `Slot`.
4. **Report empty slots** → `Inventory` notifies an observer; a telemetry client sends it.
5. **Walk-away timeout** → a timer calls `cancel()` after ~60 s through the same state machine.
6. **What did you leave out?** Card payment, network reporting, refrigeration, discounts — say they are scope decisions.

Real-world example

Cashless readers (**Nayax**, **Cantaloupe ePort**) plug in as another payment device; coin mechanisms light **"exact change only"** when a tube runs low — our planner saying it cannot make change.

In the interview

45 minutes: 5 questions → 5 entities → **10 draw the state diagram and table** → 15 code the interface, two or three states and `checkPaid`/`refundAll` → 10 edge cases.

> **30-second answer:** "The question is the order of events, so I use the State pattern: Idle, Selected, HasMoney, Dispensing, OutOfStock, all implementing an interface listing every button. The machine forwards to its state. Money goes into the box on insert, so a refund is 'make exact change for the balance'; if exact change is impossible the sale is refused and everything returned. Integer cents."

- *"Why not an enum with a switch?"* — Every new state means editing every switch; a State class is forced by the compiler to answer every button.
- *"Is greedy change correct?"* — Only for well-stocked canonical coin sets; use bounded coin-change DP.

Self-check

1. **Q:** What makes the design complete? **A:** Every state implements every button, so every (state, button) cell has an answer.
2. **Q:** Why is `restock` not on `VendingState`? **A:** Not a customer action, valid in every state (ISP).
3. **Q:** A greedy failure? **A:** 30¢ with one quarter and three dimes.

4. **Q:** How is "two items for one payment" prevented? **A:** Balance reset to 0 and Dispensing returns any coins until `take`.

TL;DR

- **State pattern is the design**: five states, one interface listing every button, no `if` on state.
- Count on the **slot**; **integer cents**; money into the **box immediately**.
- Change **exact or refuse**; greedy can fail, **DP** is correct.
- One refund path; operator actions off the state interface.

14.3 Design a Coffee Vending Machine

A **coffee machine** makes drinks (espresso, americano, latte, cappuccino, mocha, hot chocolate) from **shared ingredients** (beans, water, milk, chocolate, sugar). Unlike 14.2, a drink **consumes several ingredients at once**, and many drinks draw on the same milk. The interviewer wants: (1) **recipes as data** — a new drink is a data change; (2) **atomic, all-or-nothing deduction** — a latte takes beans, water *and* milk together or nothing, never negative even with ten concurrent orders; (3) **low-stock alerts** via Observer before an ingredient runs out.

Problem & clarifying questions

Question	Assumed answer
Drinks?	Six, loaded from a menu file.
Customisation?	Extra espresso shot (+50¢), sugar in grams.
Payment?	Amount in cents; change returned (coins covered in 14.2).
Concurrent orders?	Yes — several touch screens.
Refills?	Operator, alerted at a threshold.
Alert frequency?	**Once per crossing** of the threshold, re-armed after refill.

Requirements

Functional: (1) drinks with a **price** and a **recipe**; (2) **menu** with prices and availability; (3) selection and **payment**, **change**, full refund if the drink can't be made; (4) track inventory, deduct exactly the recipe; (5) **low-stock notifications**; (6) **refills**; (7) **customisation** without changing the menu.

Non-functional: thread-safe (no over-draw, no half recipe); data-driven; observable.

Use cases

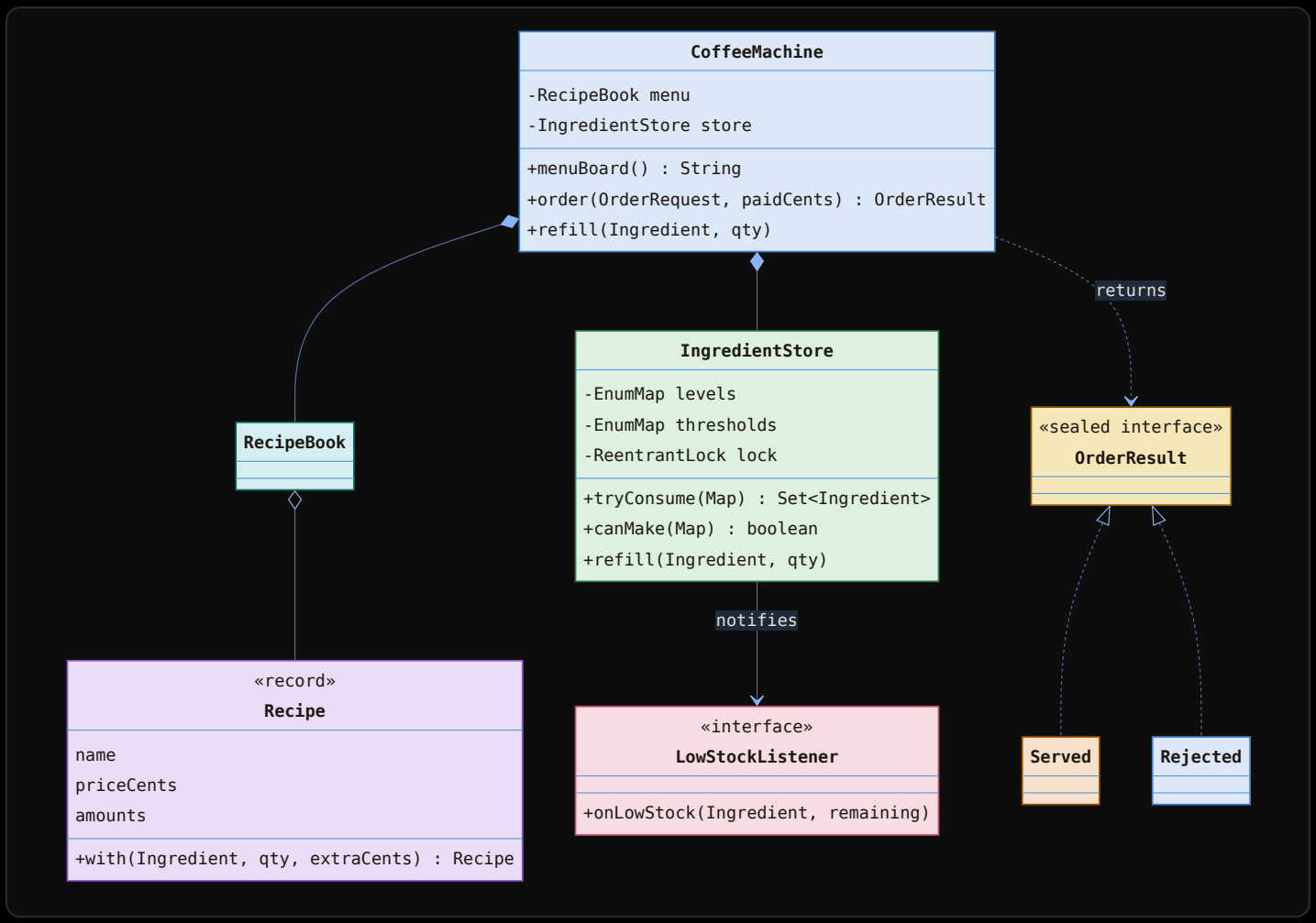
Actors: Customer (view menu, order, pay, take drink/change), Operator (refill, receive alerts), System (check stock, deduct, brew, alert). **Use cases:** view menu · order (with customisation) · pay / change / refund · refill · receive low-stock alert.

Core entities

Class	Responsibility
`Ingredient` (enum)	Beans, water, milk, chocolate, sugar — with unit (g/ml).
`Recipe` (record)	**Data**: name, price, `Map<Ingredient,Integer>`. Immutable; `with(...)` returns a customised copy.
`RecipeBook`	Parses menu text into recipes; lookup by name.
`IngredientStore`	Levels, thresholds, **one lock**; `tryConsume` = check all + deduct all atomically; alerts.
`LowStockListener`	Observer called once per threshold crossing.
`OrderRequest` (record)	Drink + extra shots + sugar grams.
`OrderResult` (sealed)	`Served(drink, price, change)` or `Rejected(reason, refund)`.
`CoffeeMachine`	Facade: menu board, `order(...)`, `refill(...)`.

Mapping to the common problem statement (*awesome-low-level-design*): `Coffee` → `Recipe`; `Ingredient` with `synchronized` updates → enum + levels in `IngredientStore`; `Payment` → `paidCents` and `OrderResult`; Singleton → injected machine. We lock the **whole recipe**, not each ingredient (per-ingredient locks can't make a multi-ingredient deduction all-or-nothing).

Class diagram



Key flows

Ordering: find recipe → apply customisation (`recipe.with(...)`) → if underpaid, `Rejected` with full refund → `tryConsume(amounts)`: lock, check **every** ingredient; if any shortage, nothing is deducted and the

order is `Rejected`; else deduct all, note thresholds crossed, **unlock, then** notify listeners → `Served` with change.

Order life cycle: `Selected → Reserved (ingredients deducted atomically) → Brewing → Served`, or `Selected → Rejected` (unknown drink, underpaid, or short) with full refund.

Design decisions & patterns used

Decision	Pattern / principle	Why	Alternative rejected
Recipes are **data** from menu text	Data-driven, OCP	New drink = new line	`Latte extends Coffee` (class explosion)
Check-and-deduct under **one lock**	Atomicity (Ch. 12)	Stops two orders both seeing "enough milk" (check-then-act)	Separate check and deduct; per-ingredient locks
All-or-nothing	Transaction semantics	No half-made recipe	Deduct one by one and roll back
`LowStockListener`	**Observer**	Operator app, dashboard, telemetry subscribe	Store calling an SMS class
Alert **once per crossing**	—	No alert storm	Alert on every order below threshold
Listeners **outside** the lock	Concurrency hygiene	Slow listener can't block orders; no deadlock on callback	Notifying inside the lock
Customisation returns a **new** `Recipe`	Immutability	Menu recipe never corrupted	Mutating it; a Decorator chain (heavier)
Sealed `OrderResult`	Algebraic data type (Java 17+/21 switch)	Callers must handle both; exhaustive `switch`	`null` or exceptions for normal refusals
No Singleton	DI	Tests build fresh machines (demo builds two)	`getInstance()`

> **Decorator or data?** *Head First Design Patterns* models add-ons as Decorators (`new Whip(new Mocha(new DarkRoast()))`). Good when add-ons have **behaviour**. Here they only change **quantities and price**, so `recipe.with(...)` is simpler. Say both and pick one for a reason.

Java implementation

Package `lld.ch14.coffeemachine`.

```java
/** A recipe is DATA: a name, a price and the amount of each ingredient. */
record Recipe(String name, int priceCents, Map<Ingredient, Integer> amounts) {
    // ...
    /** Customisation returns a NEW recipe; the menu entry is never modified. */
    Recipe with(Ingredient extra, int qty, int extraCents) {
        if (qty == 0) return this;
        Map<Ingredient, Integer> m = new EnumMap<>(Ingredient.class);
        m.putAll(amounts);
        m.merge(extra, qty, Integer::sum);
        return new Recipe(name, priceCents + extraCents, m);
    }
}

final class IngredientStore {
    // ...
    /** Returns the ingredients that are short. Empty set = success, everything was deducted. */
    Set<Ingredient> tryConsume(Map<Ingredient, Integer> need) {
        Set<Ingredient> shortages = EnumSet.noneOf(Ingredient.class);
        List<Runnable> alerts = new ArrayList<>();
        lock.lock();
        try {
            need.forEach((i, q) -> { if (levels.get(i) < q) shortages.add(i); });
            if (!shortages.isEmpty()) return shortages;              // nothing touched
            need.forEach((i, q) -> levels.merge(i, -q, Integer::sum));
            for (Ingredient i : need.keySet()) {
                int left = levels.get(i);
                if (left <= thresholds.get(i) && alerted.add(i)) {    // alert once per crossing
                    alerts.add(() -> listeners.forEach(l -> l.onLowStock(i, left)));
                }
            }
        } finally {
            lock.unlock();
        }
        alerts.forEach(Runnable::run);   // call listeners OUTSIDE the lock
        return shortages;
    }
    // ...
}
```

Full runnable code: code/lld/ch14/coffeemachine/ (see the Full Edition).

How it works: the menu is a Java text block (`Latte, 400, COFFEE_BEANS=18, WATER=40, MILK=180`) parsed by `RecipeBook.parse`. `Recipe`'s compact constructor stores an unmodifiable `EnumMap`. `refill` clears the `alerted` flag when the level goes back above the threshold. `CoffeeMachine.order` validates drink → customises → checks payment → reserves ingredients.

Output (excerpt):

```
REJECTED out of [MILK], refund 400 c
   beans left 410g (unchanged by the rejected latte)
== 10 customers order a Latte at the same instant (milk for exactly 4) ==
   ALERT to operator: MILK low, 80ml left
served=4 rejected=6 milk left=80ml (never negative)
```

Concurrency & edge cases

Situation	Risk	Handling
Ten orders, milk for four	Negative stock (check-then-act)	Check + deduct under one lock
One ingredient short	Beans taken, milk missing	Shortages found before any deduction
Slow listener (SMS)	Orders wait on the network	Listeners after `unlock()`
Many orders below threshold	Alert storm	One alert per crossing; refill re-arms
Underpayment	Drink made, money short	Payment checked before reserving
Brew fails after deduction	Pays, gets nothing	Extension: refund, optionally refill the amounts (compensation)
Menu says available, order fails	Display/order race	Menu is advisory; `tryConsume` is the truth

One lock is right: the critical section is tiny and the machine brews one drink at a time. For a cloud inventory shared by many machines, use optimistic concurrency (version / CAS on a snapshot) or a DB transaction.

Extensions & follow-up questions

1. **New drink** → one menu line. 2. **Sizes** → `recipe.scaled(size)`. 3. **Several brewers** → a `BlockingQueue` consumed by N `Brewer` threads, one store. 4. **Remote monitoring** → another `LowStockListener`. 5. **Different brewing steps** → Template Method or `Step` objects (Command). 6. **Cash** → reuse 14.2's `CashBox`.

Real-world example

The Thermoplan Mastrena (Starbucks) stores **recipes as configuration**; Starbucks and Microsoft (2019, Azure Sphere) push recipes and collect telemetry from the cloud; connected kiosks report stock remotely — the low-stock Observer.

In the interview

40 minutes: 5 questions → 5 entities → 5 "recipes are data" and the menu format → 15 code `tryConsume` and `order` → 10 concurrency and extensions.

> **30-second answer:** "Recipes are data — ingredient-to-amount maps loaded from a menu, so a new drink is a new row. The store checks and deducts **every** ingredient of a recipe under one lock: all-or-nothing, never negative. Crossing a threshold is noted and observers are notified once, after the lock is released. Orders return a sealed Served/Rejected result; rejected orders are refunded in full."

- *"Why not an `AtomicInteger` per ingredient?"* — Each is safe alone, but a recipe updates several at once; without a common lock you get partial deductions.
- *"Why notify outside the lock?"* — A slow or re-entrant listener would block all orders or deadlock.

Self-check

1. **Q:** All-or-nothing for `tryConsume`? **A:** Every ingredient deducted, or none (shortages detected first).
2. **Q:** Which race does the lock prevent? **A:** Check-then-act: two threads see enough milk and both deduct.
3. **Q:** How is an alert storm prevented? **A:** An `EnumSet` of alerted ingredients, re-armed by refill.
4. **Q:** When use Decorator for add-ons? **A:** When they carry behaviour, not just quantities.

TL;DR

- **Recipes are data**; customisation makes a new immutable recipe.
- **One lock, check-then-deduct, all-or-nothing**.
- **Observer** alerts, **once per crossing**, **outside** the lock.
- **Sealed** `OrderResult` ; full refund on rejection.

14.4 Design an ATM

An **ATM** is a cashier-less machine where a bank's customers do everyday banking outside branch hours. The interviewer checks that you keep **money correct** (never dispense without debiting, never debit without dispensing, never let two machines overdraw one account), **authenticate first**, and model the **session states** cleanly.

Problem & clarifying questions

Question	Assumed answer
Accounts / identification?	Checking and savings (daily cap); card + PIN, checked by the bank on every transaction.
Deposits available at once?	Checks **not** — they wait. Keep **total** and **available** balances.
Notes?	$100, $50, $20, $10 cassettes.
Wrong PIN?	3 wrong attempts block the card and the ATM keeps it (course leaves this out; we add it plus a journal).
Customers at a time?	One; the ATM must not shut down mid-session.

Requirements

Parts: card reader, keypad, screen, cash dispenser, deposit slot (cash/check), printer, network link to the bank.

Functional: (1) card + PIN authenticated **by the bank** first; (2) five transactions: balance inquiry, cash deposit, check deposit, withdraw, transfer; (3) the ATM **keeps the card** all session; cancel ejects it; (4) a **journal** of transactions and hardware faults; (5) a **check is held** until verified (the course holds all deposits; we credit counted cash at once); (6) **bank manager** reports.

Actors' operations: Operator — on/off key switch, refill cash, paper, ink, empty deposits. Customer — the transactions. Bank manager — total deposits/withdrawals reports, print them, check remaining cash.

Non-functional: security (no clear PINs on the ATM; limited attempts); consistency (debit and dispense together or compensate); concurrency (many ATMs, one account, no overdraft, no deadlock); auditability.

Use cases

Only Customer use cases need the PIN (Operator uses a key; Manager reads bank data). The course's activity diagrams (authenticate, withdraw, deposit check, transfer) all end by ejecting the card — **every** exit path, including declined retries, must return it.

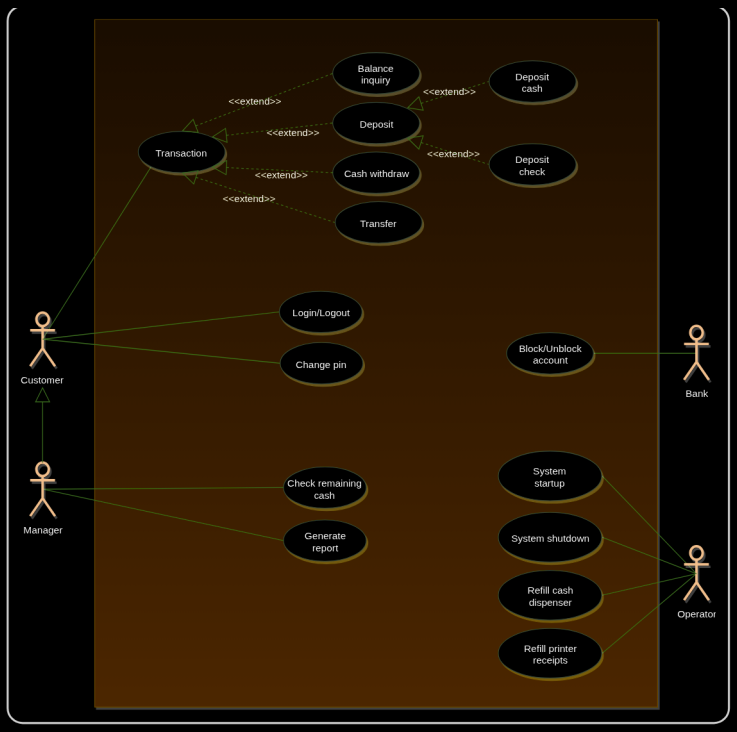


Figure — ATM use case diagram: Customer (balance, withdraw, deposit, transfer), Operator (on/off, refill cash, paper, ink, empty deposits) and Bank Manager (reports, check remaining cash) (course diagram)

Core entities

Class	Responsibility
`Card` (record)	Number and holder (course also has `Customer` with one card).
`Account` (abstract)	Id, **total** and **available** balance, a lock, withdrawal-rule hook.
`CheckingAccount`, `SavingsAccount`	Savings adds a **daily cap** — a behaviour difference, hence subclasses.
`BankService` (interface)	Authenticate, balances, withdraw, reverse, deposits, transfer.
`InMemoryBank`	Stand-in bank: PIN hashes, failed attempts, account locks, ordered-lock transfers.
`AuthResult` (sealed)	`Ok(accounts)`, `WrongPin(attemptsLeft)`, `Blocked`.
`NoteHandler` + 4 subclasses	**Chain of Responsibility** $100 → $50 → $20 → $10.
`CashDispenser`	Plans (dry run), then dispenses; can jam (fault injection).
`Transaction` (sealed) + 5 records	**Command**: one `execute(ctx)` each.
`AtmState` + 4 states	**State**: Idle → CardInserted → Authenticated, plus OutOfService.
`Atm`	Id; owns parts, session, journal; forwards buttons to state. Course hardware (`CardReader`, `Keypad`, `Screen`, `Printer`, `DepositSlot`) is folded in.

Class diagram

Course stages: (1) **money and card** — two balances (a check raises *total* now, *available* after clearing); savings overrides one method (behaviour ⇒ inheritance); (2) **the parts** — none knows what a transaction is; whether $85 can be made from loaded notes is the **dispenser's** job; the reader **keeps the card** all session; `DepositSlot` (the opening) ≠ the deposit *transaction*; (3) **five transactions**, one `execute` each; the course's `Withdraw` checks **denominations → account → dispenser** and debits only after notes are counted; (4) **the machine** owns its parts and holds the session (one customer at a time); the `Bank` owns its ATMs.

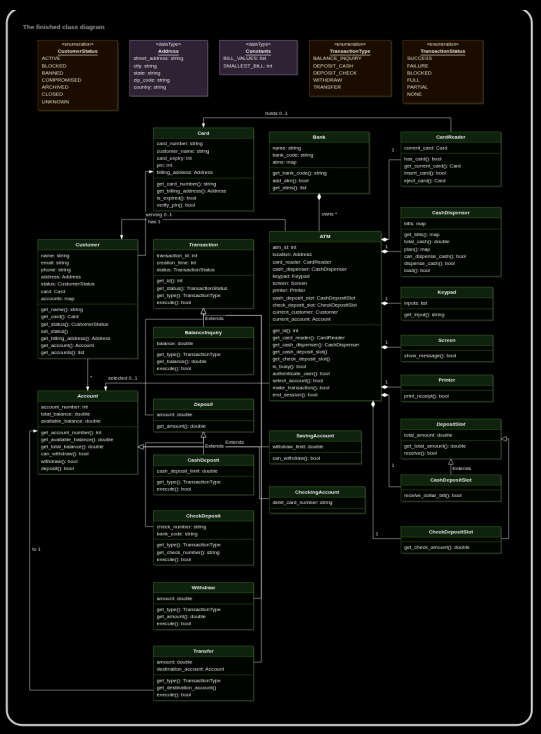
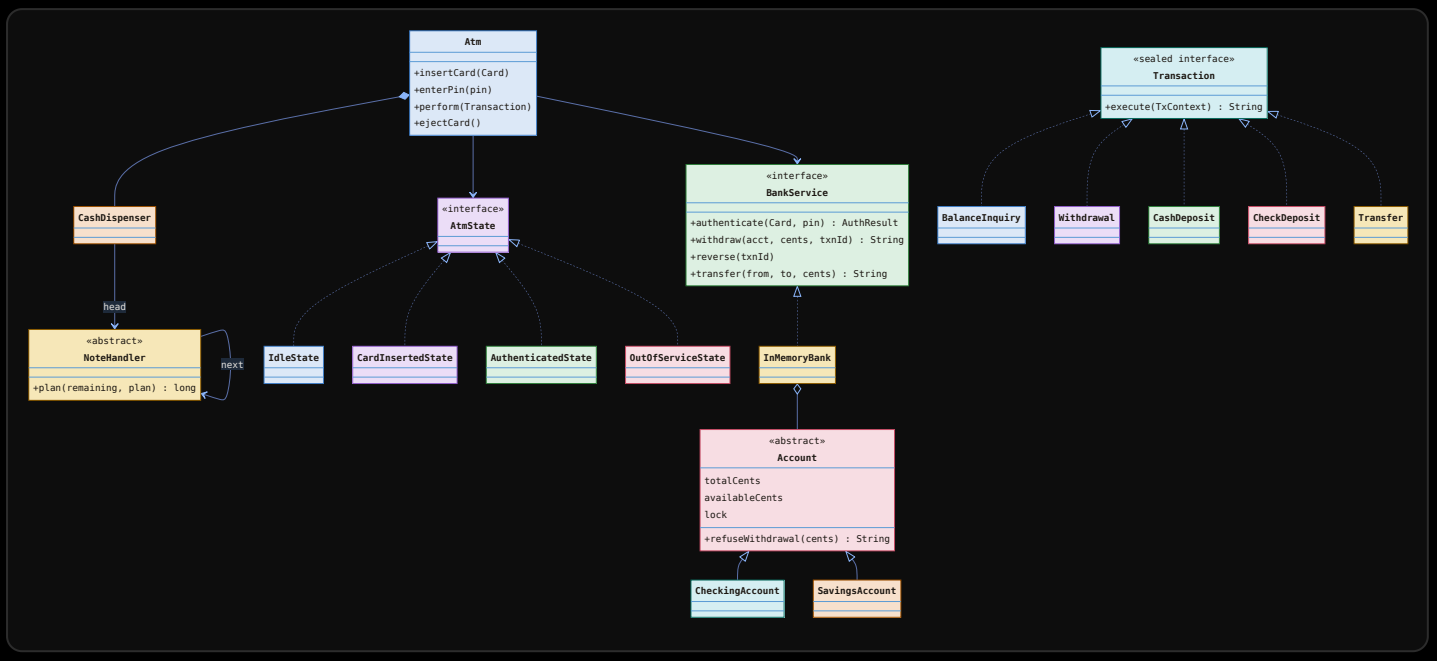


Figure — The finished class diagram of the ATM system (course diagram)



Key flows

Session states: Idle → CardInserted → Authenticated (PIN OK) → Idle (finish/cancel, card ejected). **3rd wrong PIN → Idle, card retained.** Session ends with no cash → OutOfService (card out first); key switch only when Idle; refill → Idle. **Balance inquiry:** the ATM creates a transaction, which asks the account; the result goes to the screen.

Withdrawal with compensation (our code): multiple of $10? → `plan(280)` dry run on the note chain (2×100, 1×50, 1×20, 1×10) → bank `withdraw` (lock account, check rule, debit) → `dispense(plan)` → on **jam**, `reverse(txnId)` and report "account NOT charged" → journal the result.

Design decisions & patterns used

Decision	Pattern	Why	Alternative rejected
Session as state classes	**State**	Each button means something different per state	Booleans + nested `if`s
Dispenser as chain of note handlers	**Chain of Responsibility**	Each takes what it can, passes the rest; a $5 cassette is one new handler	Hard-coded note list
Plan first, then dispense	—	Dry run touches nothing	Removing notes while searching
Each transaction = one class	**Command**	Rules in one place; can be journaled, queued, replayed	`switch (type)` in the ATM
`Transaction`, `AuthResult` sealed	Sealed types	Exhaustive `switch`	`instanceof` chains
ATM → `BankService` interface	Dependency Inversion	In-memory, ISO 8583, or a test fake	ATM changes balances
Total vs available	Domain modelling	Check holds correct	Single balance
Savings subclass	Inheritance for **behaviour**	Overrides one hook	`if (type == SAVINGS)`

> **Debit before or after dispensing?** The course debits **after** counting notes (safer for the customer) and admits a crash in between leaves the bank short. Our code debits then dispenses, with a **compensating** `reverse(txnId)` on a jam. Real systems make the pair atomic (**hold/authorise, dispense, confirm**) and **reconcile** from the journal. This is the "power fails halfway" follow-up.

> **Cash deposits.** Many banks make machine-counted cash available immediately and hold checks. `depositCash` credits both balances, `depositCheck` only the total — a one-line policy change in the bank, not the ATM.

Java implementation

Package `lld.ch14.atm`. Amounts are cents (`long`).

```
abstract class NoteHandler {
    // ...
    /** Dry run: fills 'plan', returns the amount that nobody in the chain could cover. */
    long plan(long remaining, Map<Long, Integer> plan) {
        int use = (int) Math.min(count, remaining / noteCents);
        if (use > 0) plan.put(noteCents, use);
        long left = remaining - use * noteCents;
        return (left == 0 || next == null) ? left : next.plan(left, plan);
    }
    // ...
}

record Withdrawal(String account, long cents) implements Transaction {
    static final long SMALLEST_NOTE = 10_00;

    public String execute(TxContext c) {
        if (cents <= 0 || cents % SMALLEST_NOTE != 0) return "Withdrawal refused: amount must be a multiple of
$10";
        var plan = c.dispenser().plan(cents);                                  // 1. can the notes make it?
        if (plan.isEmpty()) return "Withdrawal refused: ATM cannot make " + Money.fmt(cents) + " from its notes";
        String refusal = c.bank().withdraw(account, cents, c.txnId());          // 2. atomic debit at the bank
        if (refusal != null) return "Withdrawal refused: " + refusal;
        try {
            c.dispenser().dispense(plan.get());                                 // 3. hand out the cash
        } catch (IllegalStateException hardwareFault) {
            c.bank().reverse(c.txnId());                                        // compensate: undo the debit
            return "Withdrawal failed (" + hardwareFault.getMessage() + "), account NOT charged";
        }
        return "Dispensed " + Money.fmt(cents) + " as " + describe(plan.get());
    }
    // ...
}

final class InMemoryBank implements BankService {
    // ...
    /** Locks both accounts in a fixed (id) order, so two opposite transfers cannot deadlock. */
    @Override public String transfer(String from, String to, long cents) {
        if (cents <= 0) return "amount must be positive";
        if (from.equals(to)) return "same account";
        Account a = acc(from), b = acc(to);
        Account first = from.compareTo(to) < 0 ? a : b;
        Account second = first == a ? b : a;
        first.lock.lock();
        try {
            second.lock.lock();
            try {
                if (cents > a.availableCents) return "insufficient funds";
                a.availableCents -= cents;
                a.totalCents -= cents;
                b.availableCents += cents;
                b.totalCents += cents;
                return null;
            } finally {
                second.lock.unlock();
            }
        } finally {
            first.lock.unlock();
        }
    }
    // ...
}
```

Full runnable code: code/lld/ch14/atm/ (see the Full Edition).

How it works: the bank stores a SHA-256 hash of card+PIN and counts failures; the 3rd returns `Blocked` and the ATM **retains** the card (demo only — real banks verify an encrypted PIN block in an HSM). `withdraw` checks and debits under the account's `ReentrantLock`. `finishSession` picks Idle or OutOfService only after the card is out.

Output (excerpt):

```
   [ATM-7 screen] Dispensed $280.00 as 2x$100 + 1x$50 + 1x$20 + 1x$10
   [ATM-7 screen] Balance CHK-1: available $720.00, total $1220.00
   [ATM-7 screen] Withdrawal failed (note jam), account NOT charged
== Two ATMs, one account, same instant: $300 each from $500 ==
   cash handed out $300.00, JOINT left $200.00
== 2,000 opposite transfers in parallel (ordered locking: no deadlock) ==
   successful=2000, A+B=$2000.00
```

Concurrency & edge cases

Situation	Risk	Handling
Two ATMs, one account	Overdraft (check-then-act)	Check + debit under the **account lock**
A→B and B→A at once	**Deadlock**	Lock in a global order (account id)
Jam after debit	Charged, no cash	`reverse(txnId)` + journal
Amount not makeable	Partial dispense	Dry-run `plan` before touching the bank
Greedy chain misses a combination	$60 with one $50, three $20, no $10	OK for standard cassettes; DP planner (14.2) if asked
Many wrong PINs	Brute force	3 tries, block **at the bank**, retain card
Walks away / unlinked account	Misuse	Session timeout (extension) / `sessionAccounts` check
Cash runs out	Card trapped	Finish session, then `OutOfService`; refill → Idle
Key switch mid-session	Shuts down while serving	Refused unless idle

Extensions & follow-up questions

1. **Can't make the amount** → plan fails, refused before the bank is called.
2. **Power fails mid-withdrawal** → hold at the bank, dispense, confirm; on restart read the journal and confirm or reverse; bank reconciles.
3. **Daily limit for all accounts** → in `Account` or a policy object; ATM unchanged.
4. **Other banks' cards** → a routing `BankService` (Proxy/Adapter to the card network).
5. **Cardless** → a new state path `enterOneTimeCode`; transactions unchanged.
6. **Manager reports** → aggregate the journal; cash left = `dispenser.total()`. (Receipts / PIN change → more `Transaction` records.)

Real-world example

ATMs (NCR, Diebold Nixdorf) drive devices through **CEN/XFS** interfaces, talk **ISO 8583**, and use **ISO 9564** PIN blocks verified in HSMs; after a dispense error they reverse and reconcile from the journal.

In the interview

45 minutes: 5 parts, actors, transactions → 5 total vs available → 10 state diagram → 15 code `Withdrawal.execute` and the chain → 10 concurrency and follow-ups.

> **30-second answer:** "Parts behind interfaces, the session as a State machine, five transactions as Commands with one `execute`. The ATM never touches balances; it calls a `BankService`. Accounts have total and available balances so checks can be held. Withdrawal plans notes through a Chain of Responsibility, debits atomically at the bank, dispenses, and reverses on hardware failure. The bank locks per account; transfers lock in id order to avoid deadlock."

- *"Where is the PIN checked?"* — At the bank (really an HSM, from an encrypted PIN block).
- *"Two ATMs, joint account?"* — Check and debit together under the account's lock (or one DB transaction).

Self-check

1. **Q:** Why two balances? **A:** A check raises the total at once but must not be spendable until it clears.
2. **Q:** Safe order of debit and dispense? **A:** Neither alone; use hold + confirm (or debit + compensating reverse) and reconcile from the journal.
3. **Q:** How are transfer deadlocks avoided? **A:** Always lock both accounts in the same global (id) order.
4. **Q:** What happens after the 3rd wrong PIN? **A:** The bank blocks the card (every ATM refuses it) and this ATM retains it.

TL;DR

- **State** (session), **Command** (transactions), **Chain of Responsibility** (cassettes), `BankService` interface.
- **Total vs available** balance.
- Withdrawal = **plan → atomic debit → dispense → reverse on failure**, journaled.
- **Per-account lock**, transfers in **id order**, PIN attempts limited at the bank.

14.5 Design an Elevator System

A building's cars are run as a **group** by a shared controller. Two decisions carry the interview: **which car answers a hall call** (dispatching) and **in what order one car visits its stops** (scheduling).

Problem & clarifying questions

Question	Assumed answer
Floors and cars?	Floors 0 … N-1, several cars serving all.
Buttons?	Hall UP/DOWN per floor (no UP on top, no DOWN on ground); car panel per floor plus door open/close.
Displays?	In car: floor, direction, load. Per floor: one display per car.
Scheduling?	Serve every stop ahead before turning — **LOOK**.
Dispatching?	Nearest suitable car; swappable.
Capacity?	Overloaded car doesn't move and says so.
Maintenance, emergencies?	Out-of-service cars; fire alarm recalls all to the lobby.
Real time?	Simulated: one `step()` = one action (testable without a clock).

Requirements

Functional: (1) many floors and cars; (2–3) hall and car panels; (4–5) car and floor displays; (6) a hall call **picks one car**; (7) a car button **adds a stop**; (8) **serve stops ahead** before turning; (9) **never move with doors open**; (10) **max load** — overloaded car stays and says so; (11) idle car waits where it finished; (12) **out of service** cars get no riders; (13) *(ours)* **fire recall**.

Non-functional: efficient (no useless trips to the shaft end); no starvation; safety overrides scheduling; thread-safe; deterministic stepping.

Use cases

Actors: Rider (call, ride, choose destination), Maintenance engineer (out of / back in service), System (choose car, move, doors, displays). **Use cases:** call · choose destination · dispatch · move one floor · open/close doors · report overload · take out of service. Dispatch and move belong to the System; the rider only triggers them.

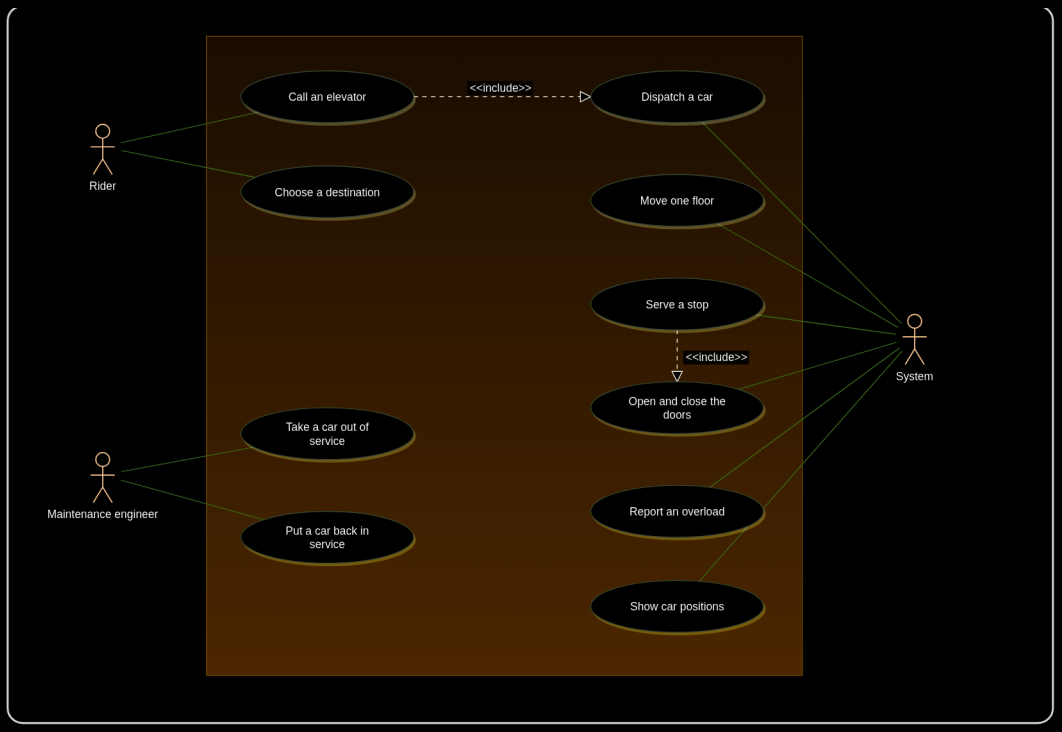


Figure — Use case diagram of the elevator system: Rider, Maintenance engineer and System (course diagram)

Core entities

Class	Responsibility
`Direction` (enum)	UP, DOWN, IDLE.
`Stop` (record)	Floor **plus** direction on leaving — "5 going up" and "5 going down" are two stops.
`StopList`	Stops in **two sorted sets** (up, down); `next(...)` implements LOOK.
`Door`	Open or closed.
`Elevator`	Floor, direction, mode, load, door, stops; `step()` does one action.
`Mode` (enum)	NORMAL, MAINTENANCE, FIRE_RECALL.
`Event` (record)	What a step did (MOVED, ARRIVED, DOORS_CLOSED, OVERLOAD, NOTHING).
`DispatchStrategy` / `NearestCarStrategy`	*Which car* answers a hall call.
`Dispatcher`	Cars + one strategy; queues calls no car can take.
`Building`	Hall lights, cars, dispatcher; ticks cars; clears lights on arrival; fire alarm.

The course also models `Button`, `HallPanel`, `ElevatorPanel`, `Display` and `Floor` one-to-one; our code folds them — say which you chose.

Class diagram

Course stages: (1) `Button` family (shared light/reset behaviour) and `Door`; (2) panels, displays, `Floor` — all **composition**; (3) **most important:** `Stop` **and** `StopList` — two sets, not one list; the next stop is reachable **without reversing**, and with nothing ahead the car heads to the **farthest** request; `Elevator` **moves**, `StopList` **chooses**; (4) `Elevator` — `step` does **one** of: close doors, refuse (overload), move one floor, stop and open; it never touches a `Floor` or another car; (5) `DispatchStrategy`, `Dispatcher`, `Building` — the only class knowing cars and floors, so it clears hall lights.

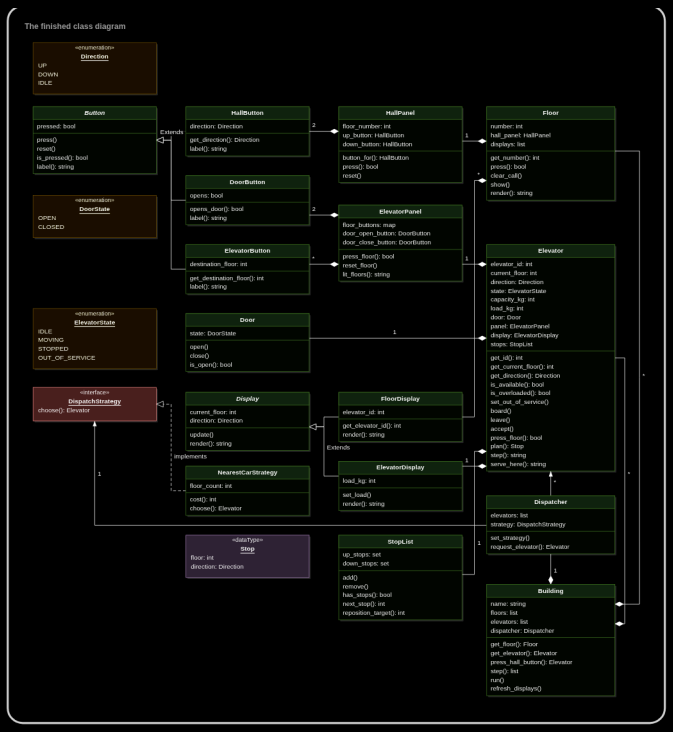
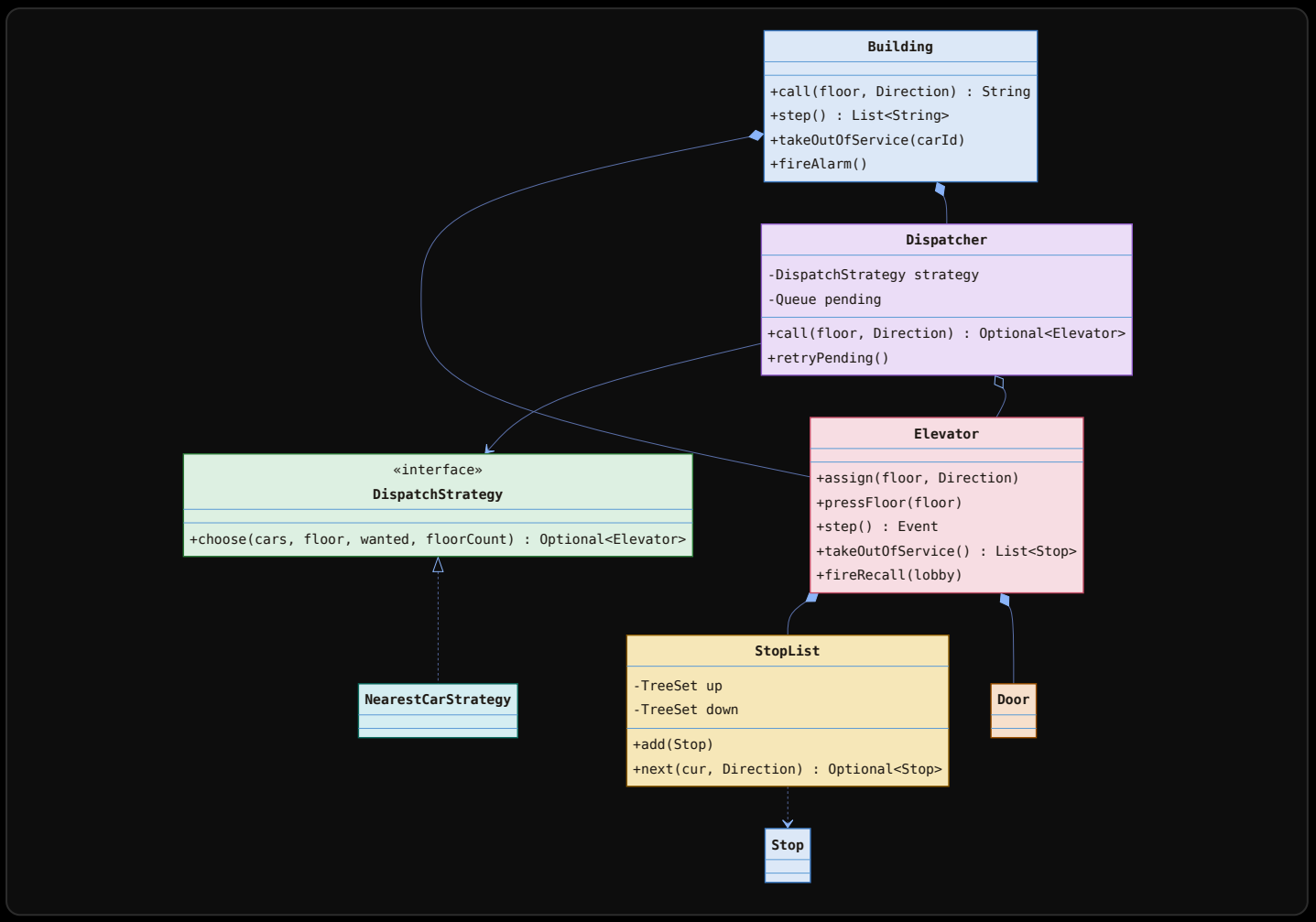


Figure — The finished class diagram of the elevator system (course diagram)



Key flows

Scheduling algorithms compared (borrowed from disk-arm scheduling — same shape: a head moving along a line of requests):

Algorithm	Rule	Good	Bad
FCFS	Arrival order	Simple	Zig-zags; very inefficient
SSTF	Nearest stop first	Short trips	**Starvation** of far floors
SCAN	Sweep to the **end of the shaft**, reverse	No starvation	Wasted travel to the ends
LOOK	Sweep only **to the last request**, reverse	No starvation, no waste	Slightly more bookkeeping
C-SCAN / C-LOOK	One direction; jump back to start	Uniform waits	Empty return trips

Our `StopList.next` (LOOK, direction-aware), going UP: (1) nearest up-stop at or above (`up.ceiling(cur)`); (2) else, if someone *above* wants DOWN, go to the **highest** such floor, then turn; (3) else turn: nearest down-stop at or below (`down.floor(cur)`); (4) else go down to the **lowest** up-stop and serve upward. DOWN mirrors it; an IDLE car picks the side of the nearest stop.

Car states: Idle → Moving → DoorsOpen (overload: stay) → Moving/Idle; Idle ⇄ Maintenance; any → FireRecall. **Hall call:** light button → strategy picks a car → `assign` → `step()` each tick → on ARRIVED, clear the light.

Dispatch cost (`NearestCarStrategy`, from the course): idle car = distance; car already coming this way **in the same direction** = distance; any other car = distance **+ two building heights**. Out-of-service cars are skipped.

Design decisions & patterns used

Decision	Pattern / principle	Why	Alternative rejected
`DispatchStrategy`	**Strategy**	Office rush vs quiet hotel; the *requirement* to swap justifies the interface	`if (officeMode)`
Sweep rule in `StopList`	SRP	Car drives, stop list decides	Next-stop logic in `step()`
Two sets (up/down)	Data modelling	"5 up" ≠ "5 down"	One list
`step()` = one action	Deterministic simulation	Tests = calls + expected events, no clock	Thread per car with `sleep`
Modes as an enum	Lightweight State	Only 3 modes; switch to State classes if modes grow	State classes from day one
Re-dispatch when a car leaves service	Robustness	Accepted calls never lost	Silently clearing stops
Queue calls when no car is free	—	"Not now" ≠ "rejected"	Dropping the call

Course's honest weak points: (1) **no threads** — a real controller locks the stop list (ours is `synchronized`); (2) the dispatcher **decides once**; (3) `Elevator` nearly does two jobs, hence `StopList`. Only **one** pattern (Strategy); out of scope: emergency calls, fire service, express cars, door sensors.

Java implementation

Package `lld.ch14.elevator`.

```java
final class StopList {
    private final TreeSet<Integer> up = new TreeSet<>();
    private final TreeSet<Integer> down = new TreeSet<>();
    // ...
    Optional<Stop> next(int cur, Direction dir) {
        if (isEmpty()) return Optional.empty();
        if (dir == Direction.IDLE) dir = nearestSide(cur);
        return Optional.of(dir == Direction.UP ? goingUp(cur) : goingDown(cur));
    }

    private Stop goingUp(int cur) {
        Integer f = up.ceiling(cur);
        if (f != null) return new Stop(f, Direction.UP);                          // work ahead
        if (!down.isEmpty() && down.last() > cur) return new Stop(down.last(), Direction.DOWN); // go up, then
turn
        f = down.floor(cur);
        if (f != null) return new Stop(f, Direction.DOWN);                        // turn around now
        return new Stop(up.first(), Direction.UP);                                // go down to lowest up-call
    }
    // ... goingDown is the mirror image
}

final class Elevator {
    // ...
    synchronized Event step() {
        if (mode == Mode.MAINTENANCE) return nothing();   // parked: never moves
        if (door.isOpen()) {
            if (mode == Mode.FIRE_RECALL && stops.isEmpty()) return nothing(); // parked at lobby, doors open
            if (loadKg > capacityKg) {
                return new Event(Event.Kind.OVERLOAD, id, floor, direction,
                        "OVERLOAD " + loadKg + "kg > " + capacityKg + "kg, doors stay open");
            }
            door.close();
            return new Event(Event.Kind.DOORS_CLOSED, id, floor, direction, "doors close");
        }
        var next = stops.next(floor, direction);
        if (next.isEmpty()) {
            direction = Direction.IDLE;
            return nothing();
        }
        Stop target = next.get();
        if (target.floor() == floor) {                      // arrived
            stops.remove(target);
            direction = stops.isEmpty() ? Direction.IDLE : target.direction();
            door.open();
            return new Event(Event.Kind.ARRIVED, id, floor, target.direction(), "doors open");
        }
        direction = target.floor() > floor ? Direction.UP : Direction.DOWN;   // never moves with doors open
        floor += direction == Direction.UP ? 1 : -1;
        return new Event(Event.Kind.MOVED, id, floor, direction, "");
    }
    // ...
}
```

Full runnable code: code/lld/ch14/elevator/ (see the Full Edition).

How it works: `TreeSet` `ceiling` / `floor` / `first` / `last` give O(log n) "nearest stop" answers. `takeOutOfService` returns the car's stops for re-dispatch; `Dispatcher.retryPending` runs every tick.

Output (excerpt):

```
== 1. LOOK on one car: at floor 4 going up; stops 6(up) 9(car) 2(down) 1(up) ==
service order [6, 9, 2, 1]  (turns at 9 and at 1: never runs to 0 or to a floor above 9)
  t15: car2@9 OVERLOAD 775kg > 680kg, doors stay open
re-dispatch hall 4 DOWN -> car2
hall 2 UP -> queued (no car available)
```

Concurrency & edge cases

Situation	Handling
Buttons pressed while the controller steps	`synchronized` car and dispatcher methods; concurrent set of hall lights
Same hall button twice	`litHallButtons.add` false: "already lit", no second dispatch
Up and down on the same floor	Two `Stop`s in two sets
Car call for current floor, doors open	Ignored
Overload	Doors stay open, OVERLOAD shown, no movement
Car out of service with accepted calls	Re-dispatched or queued, never lost
No car available	Queued; retried every tick
Fire alarm	Stops dropped, lobby added, car buttons ignored, doors open at lobby
Starvation	LOOK serves every stop within one round trip
UP on the top floor	`IllegalArgumentException`

Extensions & follow-up questions

1. **Choosing a car** → `DispatchStrategy`: zoning, **up-peak** (idle cars to the lobby at 9 a.m.), energy saving.
2. **Simultaneous presses** → synchronized dispatcher + per-car lock (the course omits it).
3. **Testing** → presses, N steps, expected events; no sleeps.
4. **Destination dispatch** → a strategy taking (from, to), grouping nearby destinations.
5. **Re-evaluate / real time** → move unserved calls to a better car each tick; a `ScheduledExecutorService` calls `step()` every second.
6. **Emergency stop / more modes** → refactor modes into State classes.
7. **Express cars / door sensor** → served-floor sets the strategy checks; an obstruction makes `step()` reopen (same slot as overload).

Real-world example

Destination dispatch (Schindler PORT, Otis CompassPlus, KONE) = extension 4; controllers switch to **up-peak** mode; fire recall follows **Phase I emergency recall** (ASME A17.1).

In the interview

45 minutes: 5 requirements/scope → 5 entities → **15 stop list and LOOK** → 10 `step()` and dispatcher → 10 concurrency, testing, follow-ups.

> **30-second answer:** "Each car keeps stops in two sorted sets, up and down, and uses LOOK: serve stops ahead, turn at the last request, never run to the shaft end for nothing. `step()` does one action — close doors, refuse if overloaded, move one floor, or stop and open — so the system is testable without a clock. A dispatcher assigns hall calls via a `DispatchStrategy` (nearest suitable car) and queues calls when no car is free. Doors, load, maintenance and fire recall override scheduling."

- *"Why LOOK, not SSTF?"* — SSTF can starve a far floor; LOOK serves everything within one round trip.
- *"Where's the lock?"* — Per car (its monitor) and around dispatch; no global lock across cars.

Self-check

1. **Q:** SCAN vs LOOK? **A:** SCAN runs to the shaft end; LOOK reverses at the last request.
2. **Q:** Car at 4 going up; stops 6(up), 9(car), 2(down), 1(up) — order? **A:** 6, 9, 2, 1.

3. **Q:** Why is scheduling in `StopList`? **A:** SRP: the car drives, the stop list decides; each testable alone.
4. **Q:** Every car in maintenance — what happens to a call? **A:** Queued, retried every tick.
5. **Q:** When would you move modes to State classes? **A:** When many modes behave differently per action.

TL;DR

- **Two sorted sets** + **LOOK** = efficient, starvation-free scheduling.
- **`step()` = one action**; doors, load, maintenance, fire recall override scheduling.
- **Strategy** for dispatch, pending queue, re-dispatch when a car leaves service.
- Per-car lock, synchronized dispatcher, concurrent hall-light set.

14.6 Design a Traffic Signal Control System

A **traffic signal controller** decides which **approaches** (roads entering: N, S, E, W) get green, for how long, and in what order. A **phase** is a set of approaches that may be green together because their paths do not cross (e.g. N and S straight). The controller cycles phases with **yellow** (warning) and **all-red** (clearance) between them. It tests a **timed state machine** with a **hard safety rule** — *two crossing flows are never green together* — plus an **emergency override** that interrupts the cycle *safely* (never cutting a yellow short).

Problem & clarifying questions

Question	Assumed answer
Roads?	Four-way (N, S, E, W); phases configurable.
Signals?	Red, yellow, green; flashing red in fault mode.
Timing?	Fixed, or adaptive from queue sensors.
Pedestrians, turn arrows?	Out of scope (extensions).
Emergency vehicles?	Green as soon as *safely* possible, held until they pass.
Many intersections?	One controller each; central coordination is an extension.
Real time?	1 tick = 1 s; tick-driven so it can be simulated.

Requirements

Functional: (1) multiple roads; (2) red/yellow/green; (3) **configurable, traffic-adaptive** durations; (4) **safe transitions**: green → yellow → all-red → next green; (5) **emergency** priority; (6) scalable/extensible; (7) fault → **all flashing red** (all-way stop).

Non-functional: safety invariant (never conflicting greens, not even for an instant); deterministic 1 s ticks; thread safety (timer vs sensor and emergency threads); observability (listeners for light changes).

Use cases

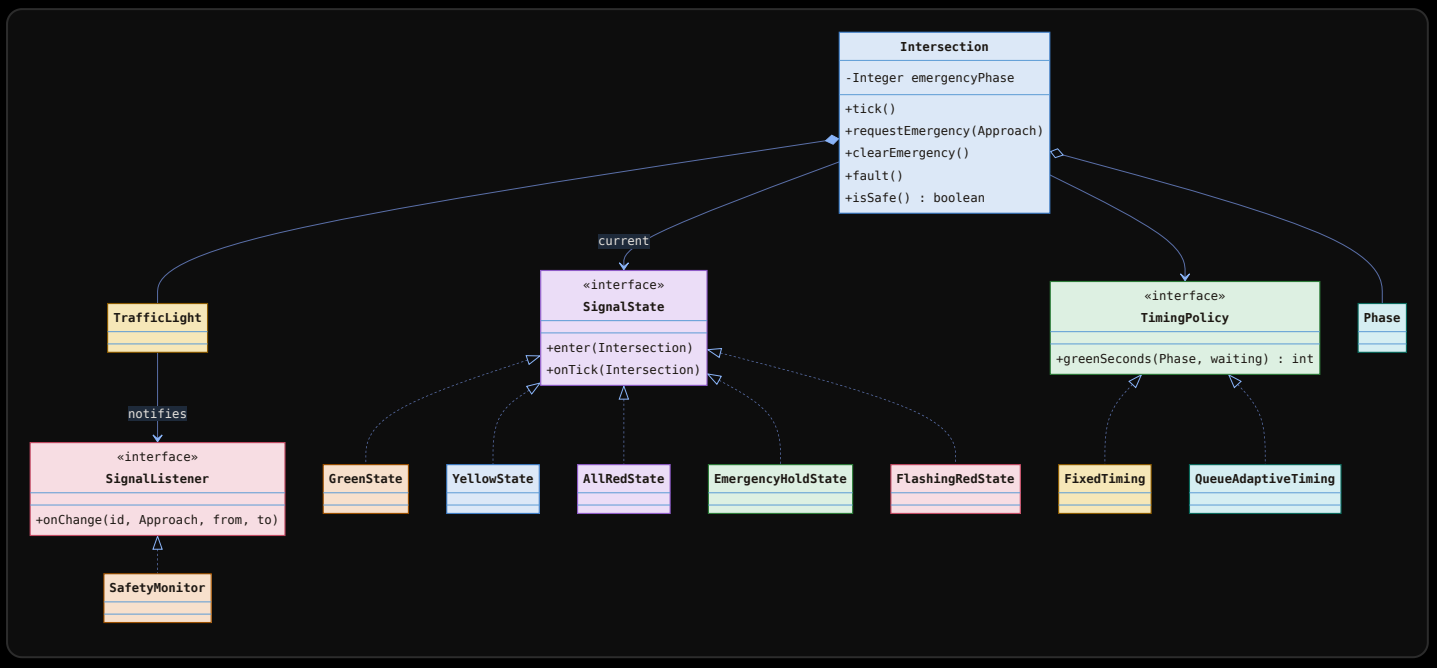
Actors → use cases: Timer → advance the cycle; Vehicle sensor → report waiting cars; Emergency vehicle → request/clear pre-emption; Traffic engineer → configure phases/timing, fault mode; Drivers see the lights.

Core entities

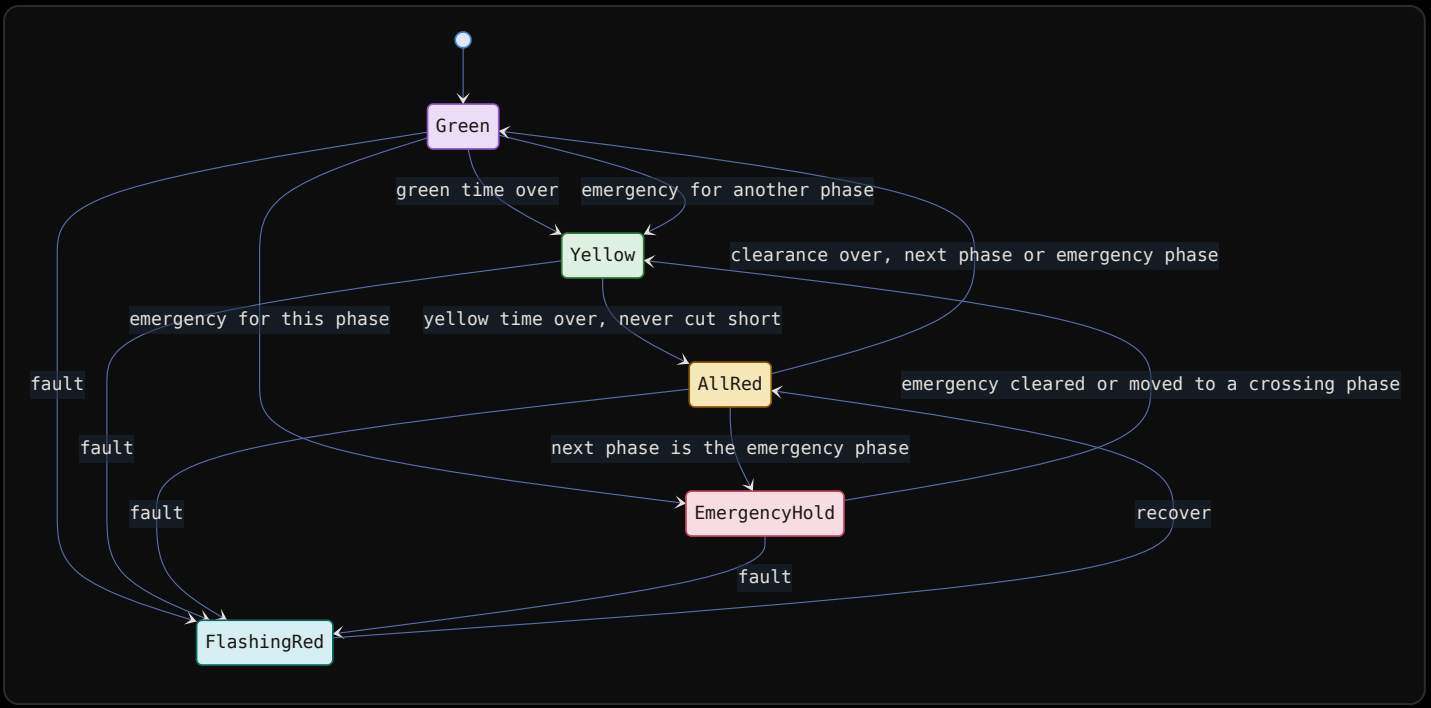
Class	Responsibility
`Signal` (enum)	RED, YELLOW, GREEN, FLASHING_RED.
`Approach` (enum)	NORTH, SOUTH, EAST, WEST.
`Phase` (record)	Name + set of approaches green together.
`TrafficLight`	One head for one approach; notifies listeners.
`SignalListener`	**Observer** for light changes.
`SafetyMonitor`	Listener checking the invariant after every change to green.
`TimingPolicy`	**Strategy**: `FixedTiming`, `QueueAdaptiveTiming`.
`SignalState` + 5 states	**State**: Green, Yellow, AllRed, EmergencyHold, FlashingRed.
`Intersection`	State context: phases, lights, timing, sensors, emergency flag; `tick()`.

Mapping to the common problem statement (*awesome-low-level-design*): `Road` → `Approach` + its `TrafficLight`; durations move into `TimingPolicy`; Singleton `TrafficController` → our `Intersection` (**not** a Singleton). We add what the reference leaves implicit: phases, all-red and the safety check.

Class diagram



Key flows



Emergency pre-emption: ambulance from WEST → `requestEmergency(WEST)` sets `emergencyPhase = East-West` → next tick N-S green ends **early** into yellow → yellow runs in full → all-red → `EmergencyHold(East-West)` → `clearEmergency()` → yellow, all-red, normal cycle.

One fixed cycle (green 5, yellow 2, all-red 1): N-S green 5 → N-S yellow 2 → all red 1 → E-W green 5 → E-W yellow 2 → all red 1 — **16 s**.

Design decisions & patterns used

Decision	Pattern	Why	Alternative rejected
Each interval a state class (`enter`, `onTick`)	**State**	Transitions and "yellow never cut short" live in one small class each	One `switch` with counters
States **carry data** (remaining seconds), new per transition	State with per-instance data	Unlike vending's stateless states, they hold a countdown	Shared singletons with counters on the context
Green time from `TimingPolicy`	**Strategy**	Fixed, adaptive, time-of-day swap at run time	Hard-coded in `GreenState`
Light changes published	**Observer**	Displays, logs, central monitoring, safety monitor	Calling each consumer
`showPhaseGreen`: others **red first**, then green	Safety by construction	Never an instant of conflicting greens	Arbitrary order
Separate `SafetyMonitor`	Defence in depth	Like hardware conflict monitors forcing flash	Trusting controller logic alone
Emergency is a **flag** states read	Safety	Only *shortens* green; yellow and all-red run in full	Jumping straight to emergency green
No Singleton controller	Rejected	Many intersections; a `TrafficControlCenter` can own them	`getInstance()`

Java implementation

Package `lld.ch14.trafficsignal`.

```java
/** State pattern: each state knows what one second of time means for it, and which state is next. */
interface SignalState {
    default void enter(Intersection x) {}
    void onTick(Intersection x);
    String name();
}

final class GreenState implements SignalState {
    // ...
    @Override public void onTick(Intersection x) {
        Integer em = x.emergencyPhase();
        if (em != null) {                                   // pre-emption: end this green early
            x.setState(em == phase ? new EmergencyHoldState(phase) : new YellowState(phase));
            return;
        }
        if (--remaining <= 0) x.setState(new YellowState(phase));
    }
    // ...
}

final class YellowState implements SignalState {
    // ...
    @Override public void onTick(Intersection x) {          // yellow is never cut short: that is unsafe
        if (--remaining <= 0) x.setState(new AllRedState(x.nextPhaseAfter(phase)));
    }
    // ...
}

final class Intersection {
    // ...
    synchronized void tick() { clock++; state.onTick(this); }
    // ...
    void setState(SignalState s) { state = s; s.enter(this); }
    // ...
    /** Turns red first, then green: never two crossing flows green, even for an instant. */
    void showPhaseGreen(int p) {
        Phase phase = phases.get(p);
        lights.forEach((a, l) -> { if (!phase.greens().contains(a)) l.set(Signal.RED); });
        phase.greens().forEach(a -> lights.get(a).set(Signal.GREEN));
    }
    // ...
    /** Safe = every approach that is green or yellow belongs to ONE phase. */
    synchronized boolean isSafe() {
        return phases.stream().anyMatch(p -> lights.entrySet().stream()
                .filter(e -> e.getValue().signal() == Signal.GREEN || e.getValue().signal() == Signal.YELLOW)
                .allMatch(e -> p.greens().contains(e.getKey())));
    }
    // ...
}
```

Full runnable code: code/lld/ch14/trafficsignal/ (see the Full Edition).

How it works: `QueueAdaptiveTiming` gives `secondsPerCar × waiting cars`, clamped to [min, max]. `setState` calls `enter` at once (set lights, load timer). `AllRedState` picks the emergency phase if pending, else the next. `SafetyMonitor` checks `isSafe()` on every change to GREEN; a real `ScheduledExecutorService` run with 20 concurrent emergency requests shows zero violations.

Output (excerpt):

```
t=37  N:G S:G E:R W:R  GREEN(0)
t=38  emergency request: WEST
t=39  N:Y S:Y E:R W:R  YELLOW(0)
t=41  N:R S:R E:R W:R  ALL_RED
t=42  N:R S:R E:G W:G  EMERGENCY_HOLD(1)
t=50  N:F* S:F* E:F* W:F*  FLASHING_RED
light changes seen by observer: 48, safety violations: 0
```

With 12 cars waiting E-W, the adaptive policy gives a 10 s green (the maximum).

Concurrency & edge cases

Situation	Handling
Emergency between ticks	Stored under the lock; next tick acts
Emergency during yellow	Yellow completes; all-red then picks the emergency phase
Emergency for the phase already green	Straight to `EmergencyHold` — no needless yellow
Two emergencies from crossing directions	Latest wins; the hold exits via yellow + all-red. Real systems queue by priority and arrival
Emergency never cleared	Maximum hold time
Timer and sensor threads overlap	`synchronized` methods; `CopyOnWriteArrayList` listeners
Adaptive timing starves a quiet road	Minimum green; every phase still gets its turn
Bug causes conflicting greens	`SafetyMonitor` detects; real controllers force flashing red
Timer drift	`scheduleAtFixedRate`; real deployments use the hardware clock

Extensions & follow-up questions

1. **Pedestrians** → WALK during the phase's green, flashing DON'T WALK before yellow, push-button requests.
2. **Left-turn arrows** → more phases; phases are data, state machine unchanged.
3. **Green wave** → a `TrafficControlCenter` sets cycle length and **offset** per intersection.
4. **Emergency corridor** → the centre pre-empts intersections along the route.
5. **Night mode** → time-of-day `TimingPolicy`, or flashing yellow/red states.
6. **Testing** → call `tick()` N times, assert snapshots.

Real-world example

SCATS and **SCOOT** adapt timing city-wide from sensors; **emergency pre-emption** emitters still time out yellow and red first; **conflict monitors / MMUs** force flash on conflicting greens — our `SafetyMonitor`.

In the interview

40 minutes: 5 requirements → 5 entities → 10 state machine → 15 code Green/Yellow/AllRed and `tick()` → 5 concurrency and follow-ups.

> **30-second answer:** "Phases are sets of approaches that can be green together; the controller is a timed State machine — Green, Yellow, All-Red, plus Emergency-Hold and Flashing-Red. A timer calls `tick()` each second; each state counts down and names the next. Green time comes from a swappable timing Strategy. An emergency only shortens green: yellow and all-red always run in full, then the emergency phase is held until cleared. Lights publish to observers, including a safety monitor that checks no crossing flows are ever green together."

- *"Why not jump straight to emergency green?"* — Cars in the junction need yellow and all-red to clear it.
- *"Why not a Singleton?"* — One per intersection; a city has thousands.

Self-check

1. **Q:** What is a phase? **A:** Approaches that may be green together because their paths don't cross.
2. **Q:** Why all-red? **A:** To let vehicles still in the junction clear it.
3. **Q:** Emergency during yellow? **A:** Yellow finishes; all-red then picks the emergency phase.
4. **Q:** Why do these states hold data but vending states don't? **A:** Each interval has its own countdown.

TL;DR

- **Timed State machine**: Green → Yellow → All-Red → next Green; plus Emergency-Hold and Flashing-Red.
- **Safety first**: red before green, yellow never cut short, conflict-monitor observer.
- **Strategy** for timing, **Observer** for lights, emergency as a flag.
- Tick-driven and synchronized: deterministic tests.

14.7 Design a Shipping Locker System (Amazon Locker)

Also asked as: "Design Amazon Locker" or "Design a parcel/pickup locker".

A **parcel locker** is a wall of numbered doors in a shop or lobby: a courier puts a parcel behind one door, the customer types a **pickup code** and that door opens (Amazon Hub Locker, InPost Paczkomat). It looks like the parking lot — hand out limited spaces of several sizes — plus three extras interviewers probe: **deadlines** (a reservation that times out, a parcel that expires), **a secret** (the code, hard to guess, never shown to the courier) and **a network** (station full → suggest another).

Problem & clarifying questions

Question	Assumed answer
Door sizes?	Small, medium, large; a parcel may be upgraded to a bigger free door.
When is the door chosen?	In advance: the carrier **reserves** it; the hold lasts 2 h, then the door is released.
How does the customer open it?	6-digit **one-time code** by SMS/app on deposit; the courier never sees it.
How long can a parcel stay?	3 days; then it **expires** and the courier takes it back.
Code guessing?	Random, single-use; 5 wrong codes in a row lock the keypad for 15 min.
Many stations?	Yes; offer the nearest one with space. Cash on delivery is an extension.
Returns?	Yes: a door is held 2 days, a single-use **drop-off code** opens it empty once, the courier collects later.

Requirements

Functional: (1) numbered doors of fixed sizes; (2) **reserve** the best-fitting free door, all-or-nothing, with a hold deadline; (3) **deposit** → one-time code **sent to the customer only**; (4) **pickup** opens one door once; (5) **expiry** → customer told, courier collects for return; (6) keypad lockout; (7) nearest station with space; (8) doors out of service; (9) **customer return** to a locker with a drop-off code.

Non-functional: never one door for two parcels under concurrency; codes unpredictable, single-use, stored as a hash; deadlines correct even if the timer is late; clock and randomness injected.

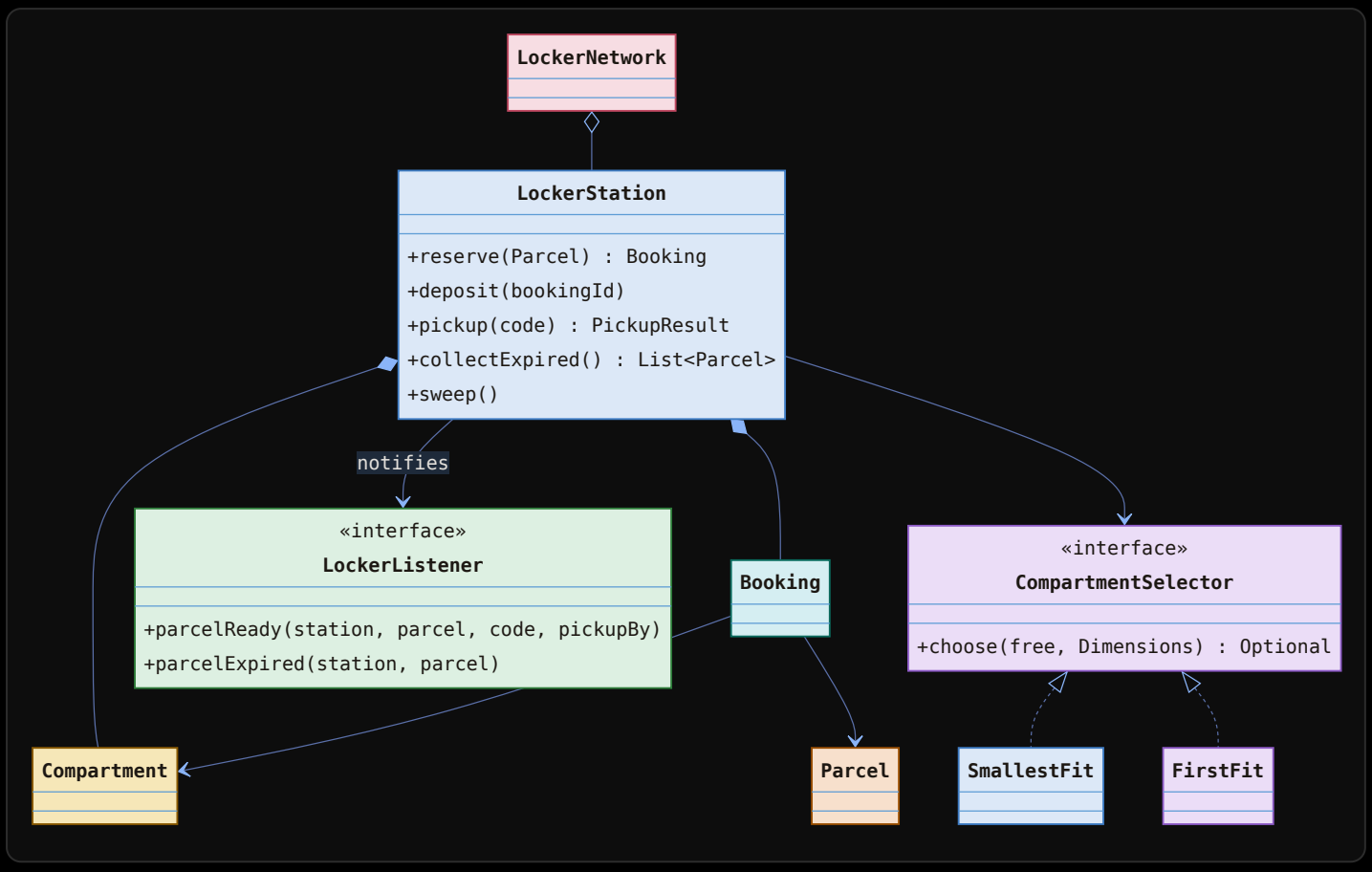
Use cases

Actors → use cases: Customer → find a locker near me, pick up with code, start a return, drop off with code; Carrier system → reserve; Delivery agent → deposit (includes: send code), collect expired parcels and returns; Locker admin → door out of service; Scheduler → sweep deadlines and notify. Keypad lockout *extends* pickup after 5 wrong codes. (Use-case and activity diagrams: Full Edition.)

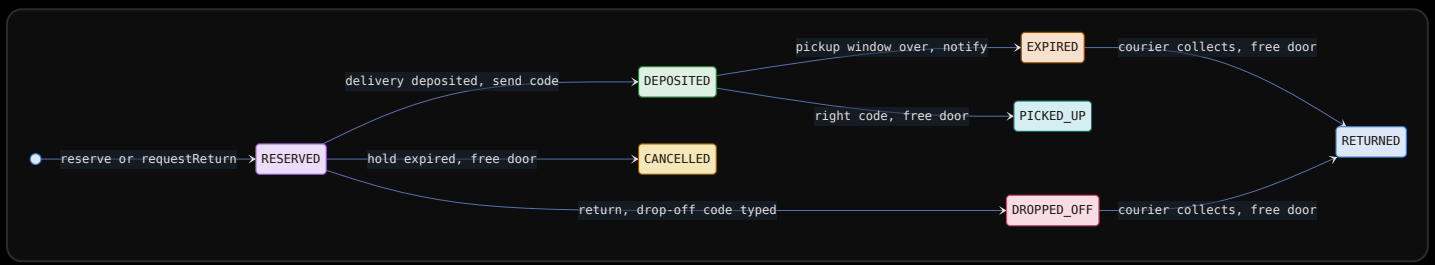
Core entities

Class	Responsibility
`Size` (enum), `Dimensions`, `Parcel` (records)	Door sizes with `fits()` (sort both boxes' sides, compare in order); parcel data.
`Compartment` + `CompartmentState`	One door: FREE, RESERVED, OCCUPIED, OUT_OF_SERVICE; legal moves in `moveTo`.
`Booking` + `BookingKind` + `BookingStatus`	One parcel's stay, DELIVERY or CUSTOMER_RETURN (statuses below); deadlines and code **hash**.
`PickupResult` (sealed)	`DoorOpened`, `DropOffDoorOpened`, `WrongCode`, `KeypadLocked`, `ParcelExpired`.
`CompartmentSelector`	**Strategy**: `SmallestFit`, `FirstFit`.
`LockerListener`	**Observer**: parcel ready (with code), parcel expired.
`LockerStation`, `LockerNetwork`	Doors, bookings, code index, lockout under one lock; nearest station with space.

Class diagram



Key flows



Delivery: carrier `reserve(parcel)` → `SmallestFit` picks the smallest free door that fits → door RESERVED, hold 2 h → courier `deposit` → door OCCUPIED, random code, only its SHA-256 hash stored → listener sends the code to the customer **after the lock is released** → customer `pickup(code)` → hash matches → door FREE, code deleted. The door has its own small machine: FREE ↔ RESERVED → OCCUPIED → FREE, and FREE ↔ OUT_OF_SERVICE.

Station full / courier late: no door fits → ask the network for the nearest station with space (none → home delivery); courier misses the 2 h hold → booking CANCELLED, reserve again.

Keypad (pickup or return drop-off): locked? → show wait. Hash unknown? → count it; the 5th in a row locks for 15 min. Delivery expired? → "will be returned". Otherwise the booking's kind decides: delivery → door opens, PICKED_UP, door FREE; return → empty door opens, DROPPED_OFF, door OCCUPIED. Either way the code is deleted. The courier's visit (`collectExpired`, `collectReturns`) marks both kinds RETURNED and frees the doors.

Design decisions & patterns used

Decision	Pattern	Why	Alternative rejected
Door choice as `CompartmentSelector`	**Strategy**	Smallest-fit saves big doors; first-fit, eye-level-first swap in	`if (mode == …)` in `reserve`
Door state separate from booking status	Two enum state machines	Doors are reused; bookings keep deadlines and history	One field mixing both
Sealed `PickupResult`	Sealed result types	Every keypad outcome handled, compiler-checked	`null` / magic strings
Random, single-use, hashed codes	Security by design	A DB leak reveals no live code; courier never sees it	Plain codes; codes derived from the parcel id
Listeners called outside the lock	**Observer**	A slow SMS gateway can't block the keypad	SMS call inside `deposit`
Lazy expiry on every call + `sweep()` for messages	Check deadlines on read	Doors come back even if the scheduler is late	A reaper thread as the only mechanism
Injected `Clock`, `RandomGenerator`	DI	Deterministic tests	`Instant.now()`, `new Random()` inside
One lock per station	Monitor	Tens of doors, microsecond calls	Per-door CAS (needless here)

Java implementation

Package `lld.ch14.locker`.

```java
// File: lld/ch14/locker/LockerStation.java
    synchronized Booking reserve(Parcel parcel) {
        expireOverdue();
        // ... reject a parcel that already has an active booking
        List<Compartment> free = compartments.values().stream().filter(Compartment::isFree).toList();
        Compartment c = selector.choose(free, parcel.dims())
                .orElseThrow(() -> new LockerException("No free compartment fits " + parcel.dims() + " at " +
id));
        c.moveTo(CompartmentState.RESERVED);
        Booking b = new Booking(id + "-B" + nextBooking++, parcel, c, now() + reservationHold.toSeconds());
        bookings.put(b.id, b);
        return b;
    }

    void deposit(String bookingId) {
        Booking b;
        String code;
        synchronized (this) {
            // ... expireOverdue(); booking must be RESERVED
            b.compartment.moveTo(CompartmentState.OCCUPIED);
            b.status = BookingStatus.DEPOSITED;
            b.pickupBy = now() + pickupWindow.toSeconds();
            do { code = "%06d".formatted(random.nextInt(1_000_000)); }       // unique among active codes here
            while (bookingByCodeHash.containsKey(hash(code)));
            b.codeHash = hash(code);
            bookingByCodeHash.put(b.codeHash, b.id);
        }
        for (LockerListener l : listeners) l.parcelReady(id, b.parcel, code, b.pickupBy);   // outside the lock
    }

    synchronized PickupResult pickup(String code) {
        expireOverdue();
        long t = now();
        if (t < lockedUntil) return new KeypadLocked(lockedUntil - t);
        String bookingId = bookingByCodeHash.get(hash(code));
        if (bookingId == null) {
            // ... count the wrong code; the 5th in a row sets lockedUntil = t + 15 min and returns KeypadLocked
            return new WrongCode(MAX_WRONG_CODES - wrongCodes);
        }
        wrongCodes = 0;
        Booking b = bookings.get(bookingId);
        if (b.status == BookingStatus.EXPIRED) return new ParcelExpired(b.parcel.id());
        b.status = BookingStatus.PICKED_UP;
        b.compartment.moveTo(CompartmentState.FREE);
        bookingByCodeHash.remove(b.codeHash);                        // a code works once
        return new DoorOpened(b.compartment.id(), b.parcel.id());
    }
    // ... expireOverdue(): RESERVED past hold -> CANCELLED (door FREE); DEPOSITED past window -> EXPIRED
```

Full runnable code: code/lld/ch14/locker/ (see the Full Edition).

How it works: `Size.fits` sorts the sides of box and door and compares in order (no need to try six rotations). `sweep()` (called by a scheduler) runs the same deadline check and sends "not collected" messages outside the lock. `LockerNetwork.nearestWithSpace` sorts stations by distance and returns the first within the radius that can fit the parcel now.

Output (excerpt):

```
P3 9x12x12cm    -> MG01-03(MEDIUM)
error: Parcel P1 already has a locker
SMS to +91-98450-00001: parcel P1 is in locker MG01, code 694696, collect by 2026-10-04T09:00:00Z
right code  -> DoorOpened[compartmentId=MG01-01, parcelId=P1]
same code again -> WrongCode[attemptsLeft=4]  (a code works once)
guess 000004 -> KeypadLocked[secondsLeft=900]
booking MG01-B4 is CANCELLED, free now: {SMALL=2, MEDIUM=1, LARGE=1}
customer types the code -> ParcelExpired[parcelId=P3]
station F: small parcel -> F-01(LARGE), then large parcel -> REJECTED
station S: small parcel -> S-02(SMALL), then large parcel -> S-01(LARGE)
reserved=20 rejected=30 distinct doors=20
customer types the drop-off code -> DropOffDoorOpened[compartmentId=RT04-02, parcelId=R1]
2 days later, never dropped off: booking RT04-B2 is CANCELLED, old code -> WrongCode[attemptsLeft=4]
```

The mug was upgraded to MEDIUM because both small doors were taken; `FirstFit` wasted the large door and had to refuse a large parcel; 50 concurrent reservations on 20 doors gave exactly 20 distinct doors.

Concurrency & edge cases

Situation	Handling
Two carriers, last door	`synchronized reserve`: choose + mark is one step
Courier late past the hold	Booking CANCELLED; `deposit` refuses; carrier re-reserves
Retried reserve for the same parcel	Rejected while a booking is active (idempotent by parcel id)
Duplicate random code	Regenerate until unique among live codes
Guessing	10^6 codes, lockout after 5 wrong, codes expire
Scheduler down	Lazy expiry still frees doors; only the SMS is late
`canFit` true but `reserve` fails	`nearestWithSpace` is a hint; reserve, then try the next station
Broken door	Only a FREE door can go OUT_OF_SERVICE (`moveTo`)

Extensions & follow-up questions

1. **Extend pickup window** → move `pickupBy` once, only while DEPOSITED.
2. **Cash on delivery** → a payment step before the door opens.
3. **Keep large doors for large parcels** → a selector that refuses upgrades when few large doors are left.
4. **Thousands of stations** → shard by station; geo index (geohash, R-tree) for the network search.
5. **Database** → `UPDATE compartment SET state='RESERVED' WHERE id=? AND state='FREE'` and check one row changed.

Real-world example

Amazon Hub Locker sends a pickup code and returns uncollected parcels after a limited number of days (our EXPIRED → RETURNED path); InPost's Paczkomat lockers in Poland open with a code or an app and are dense enough that "nearest locker with space" matters. In both, the system — not the courier — chooses the door.

In the interview

40 minutes: 5 questions → 5 entities → 8 the two state machines → 15 code `reserve` / `deposit` / `pickup` → 7 security, concurrency, follow-ups.

30-second answer: "Doors come in a few sizes; a Strategy picks the smallest free one that fits, and choose-and-mark runs under the station lock so two carriers never get the same door. On deposit I generate a random single-use code, store only its hash and notify the customer through an observer after releasing the lock. Pickup matches the hash, frees the door and deletes the code; five wrong codes lock the keypad. Holds and pickup windows are checked on every call, so doors come back even if the scheduler is late."

- *"Why a Booking separate from the Compartment?"* — The door is reused; the booking keeps each parcel's deadlines and history.
- *"Why hash the code?"* — Anyone reading the database or logs could otherwise open doors.

Self-check

1. **Q:** How does `fits` handle a box on its side? **A:** Sort both sets of sides and compare in order.
2. **Q:** What frees a door if the scheduler never runs? **A:** Lazy expiry at the start of every call.
3. **Q:** Why notify after the `synchronized` block? **A:** A slow SMS call must not hold the station lock.
4. **Q:** Which pattern swaps smallest-fit for first-fit? **A:** Strategy (`CompartmentSelector`).

TL;DR

- **Strategy** picks the smallest free door that fits; choose-and-mark is **atomic** per station.
- Two **state machines**: door and booking (RESERVED → DEPOSITED → PICKED_UP / EXPIRED → RETURNED / CANCELLED; returns RESERVED → DROPPED_OFF → RETURNED).
- Codes **random, single-use, hashed**, sent via an **Observer** outside the lock; lockout after 5 wrong.
- Deadlines enforced **lazily**; inject clock and randomness.

Chapter summary

Problem	Core difficulty	Key data structure	Key patterns	Concurrency tool
Parking lot	Allocate spots without double booking	Concurrent sorted set of free spots per type	Strategy, Factory, Observer	`pollFirst` + CAS, per-ticket lock, `putIfAbsent`
Vending machine	Legal order of actions; exact change	`EnumMap` of coin counts; DP table	State	`synchronized` machine
Coffee machine	All-or-nothing multi-ingredient deduction	`EnumMap` levels, recipes as data	Observer, sealed results	One `ReentrantLock`; notify outside the lock
ATM	Money correctness and authentication	Total vs available balance; note chain	State, Chain of Responsibility, Command	Per-account lock; ordered locking for transfers
Elevator	Scheduling and dispatching	Two `TreeSet`s (up/down stops)	Strategy (+ modes)	Per-car lock, synchronized dispatcher
Traffic signal	Timed transitions with a safety invariant	Phases as data	State, Strategy, Observer	Synchronized tick; scheduled executor
Shipping locker	Best-fit door, holds and pickup windows, secret codes	Doors per station; bookings; code-hash index	Strategy, Observer, enum state machines, sealed results	One lock per station; lazy expiry; notify outside the lock

Chapter self-check

1. **Q:** Stateless vs data-carrying states? **A:** Vending and ATM share stateless singletons; traffic-signal states hold countdowns, new per transition.
2. **Q:** The check-then-act race in parking, coffee, ATM? **A:** Two gates one spot (atomic `pollFirst`/CAS); two orders the last milk (one lock around check+deduct); two ATMs one account (check+debit under the account lock).
3. **Q:** Where did we reject a Singleton? **A:** Parking lot, coffee machine, traffic controller — hidden global, untestable, and real deployments have many.
4. **Q:** Where does greedy fail? **A:** Change with limited coins (vending) and notes with limited cassettes (ATM); use bounded coin-change DP.

5. **Q:** Which designs are tick-driven and why? **A:** Elevator and traffic signal — deterministic, clock-free tests.
6. **Q:** Which case enforces deadlines lazily? **A:** The shipping locker — every call expires overdue holds/parcels first; the sweep only sends messages.

Chapter TL;DR

- Find the **one hard method** per problem and spend your time there.
- **State** for machines whose buttons change meaning; **Strategy** for changing rules; **Observer** for boards, alerts, monitors; **Chain of Responsibility** and **Command** for the ATM.
- Make every **check-then-act** atomic, lock at the **smallest** scope, lock in a **fixed order**, call listeners **outside** locks.
- Prefer injected instances over Singletons, integer cents over doubles, tick-driven logic over sleeps.
- Deadlines (holds, pickup windows): check on every call; secrets (pickup codes): hashed, single-use.